Grade 9 Physics

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

FLAGGED-PARTIAL BUILD: Units 1-11 complete. Unit 12 exercise + Units 13-14 unobtainable (see below).

Units solved in this PDF (every Questions section: multiple choice, brief answers and numerical problems where the book prints them):

Unit 1: Introduction
Unit 2: Physical Quantities
Unit 3: Kinematics
Unit 4: Dynamics and Forces
Unit 5: Momentum
Unit 6: Turning Effect of Forces
Unit 7: Centripetal Force
Unit 8: Work, Energy and Power
Unit 9: Pressure
Unit 10: Deformation of Solids
Unit 11: Density
Unit 12: Particle Theory of Matter — LESSON ONLY in source (pp.232-234); exercise UNOBTAINABL
Unit 13: Magnetism — NOT IN SOURCE FILE; UNOBTAINABLE
Unit 14: Space Physics — NOT IN SOURCE FILE; UNOBTAINABLE

Source: BTBB Grade 9 Physics textbook file, 235 pages. Medium confirmed from the book itself: cover, Contents (PDF pp.4-10) and all units are in English; book page = PDF page. Exercises were read from rendered pages (book pp.19-20, 50-53, 79-83, 108-110, 121-122, 140-141, 149-151, 181-183, 202-204, 214-215, 229-231). Questions are reproduced from each unit's Questions section; answers are written in medium length, and numerical problems show every step. FLAGGED-PARTIAL BUILD — READ FIRST: the BTBB source file for this book is TRUNCATED. It has only 235 pages (96,528,336 bytes; p.235 blank) while the book's contents run to book p.307. This PDF therefore solves Units 1-11 completely (every Questions section: multiple choice, brief answers and numerical problems). Unit 12 (Particle Theory of Matter, book p.232) is present in the file only as lesson pages 232-234; its end-of-unit exercise is NOT in the file. Units 13 (Magnetism, book p.250) and 14 (Space Physics, book p.287) are NOT in the file at all. The Unit 12 exercise and Units 13-14 are flagged here as UNOBTAINABLE and have NOT been invented, reconstructed or borrowed from any other book. A complete second source was searched for on 2026-10-26 (BTBB's own portal serves the identical 235-page file; Taleem360's copy is the same truncation; Tayarri returns a 404) and none was found.

If a complete BTBB Physics 9 file becomes available, this PDF must be completed with it. Book quirks noted honestly (answers follow the book's intended option unless a correction is stated; see the manifest for detail): Unit 2 MCQ iv has no true base quantity among its options — (a) Distance is ticked as the closest (length) option. Unit 2 Numerical 6 mislabels its third part as (d); it is answered as (c). Unit 2 Numerical 11's printed answer line is garbled (four entries for five parts, (c) repeated); the counts given follow the significant-figure rules. Unit 3 Numerical 15's stated 5 s and 80 m are inconsistent with each other; the answer follows the stated time, as printed. Unit 4 MCQ viii has NO correct option (10 kg weighs 98 N on Earth, not listed); (b) 9.8 N is ticked only as the closest printed value and the correct 98 N is stated. Unit 4 Numerical 2's printed answer (10 kg) is wrong; 980 / 9.8 = 100 kg. Unit 4 Numerical 5(c)'s printed answer (40 m) is wrong; the correct distance is 50 m. Unit 5 is headed 'Chapter 5' in the book although the contents list it as Unit 5. Unit 6 is headed 'Tunning Effect of Forces' in the book; the correct word is Turning. Unit 6 Numerical answers print the torque unit as 'N s'; the correct unit is N m. Unit 7 MCQ iii prints options (c) and (d) identically (mv^2 / r); (c) is ticked. Unit 7 Numerical 3's printed answer (2.38 x 10^20 N) is wrong by a factor of about 230; the book's own data give about 5.5 x 10^22 N. Unit 9 Numerical 1(d)'s printed unit (N) is wrong; the value 15,484 is a pressure, in Pa. Unit 9 Numerical 3's printed maximum (4940 N) is wrong in value and unit; the correct maximum is 2940 Pa (minimum 735 Pa is correct). Unit 9 Numerical 8's wording is ambiguous about whether 0.1 m^2 is one foot's area or all four feet; the book's answer (10,000 Pa) follows the per-foot reading. Unit 11 MCQ iii mislabels its fourth option as (c) again; it is the g cm^-3 option and is ticked. Unit 11 Numerical 2's printed answer (5.56 x 10^3 kg m^-3) fits a radius of about 6.37 x 10^6 m, not the printed 6.27 x 10^6 m; the printed data give about 5.81 x 10^3 kg m^-3. Unit 11 Numerical 5(b)'s printed exponent is wrong by 1000 (1 mm^3 = 10^-9 m^3, so 4.29 x 10^-8 m^3). Unit 11 Numerical 6(b) has garbled units and a wrong exponent (correct: 2.06 x 10^-7 m^3) and its printed (c) of 0.103 kg is 1000 times too large (correct: 1.03 x 10^-4 kg = 0.103 g). Unit 11 Numerical 7's printed volume (591 cm^3) and mass (6.86 kg) do not follow from its printed diameters (recomputed: 584 cm^3, 6.78 kg), and its printed iron density (11600 kg m^-3) is far above the usual value of about 7900 kg m^-3 but is used as printed. The book's Sample Problem 11.11 states a blood mass of 53 g but solves with 5.3 g; the 5.3 g reading is the only one consistent with its answer (1.06 g cm^-3).

Unit 1: Introduction

Questions 1

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) The study of matter, energy and their mutual interactions is called:

(a) Mechanics

(b) Optics

(c) Heat

(d) Physics ✓

(ii) The branch of Physics that deals with conversion of heat energy into other forms of energy (i.e., mechanical, electrical or chemical energy etc.) or vice versa is called:

(a) Mechanics

(b) Atomic Physics

(c) Thermodynamics ✓

(d) Acoustics

(iii) The study of structure and properties of atoms is called:

(a) Thermodynamics

(b) Optics

(c) Atomic Physics ✓

(d) Nuclear Physics

(iv) The branch of Physics that deals with study, design and use of devices based on behaviour of electrons in semiconductors is called:

(a) Quantum Mechanics

(b) Plasma Physics

(c) Solid State Physics

(d) Semiconductor Physics ✓

(v) It deals with the study of production and properties of ionic state of matter.

(a) Atomic Physics

(b) Nuclear Physics

(c) Solid State Physics

(d) Plasma Physics ✓

(vi) It deals with the study of Physics involved in the study of universe i.e., the properties and interaction of celestial objects.

(a) Geophysics

(b) Biophysics

(c) Astrophysics ✓

(d) Plasma Physics

(vii) The study of internal structure of the Earth is called:

(a) Atomic Physics

(b) Nuclear Physics

(c) Computational Physics

(d) Geophysics ✓

(viii) The branch of Astronomy that deals with the study of the origin, evolution and structure of the Universe is called:

(a) Astrophysics

(b) Climate Physics

(c) Cosmology ✓

(d) Plasma Physics

B. Give brief answers to the following questions.

(i) Define science. What are its different branches?

Science is the study of natural phenomena and the laws governing them, built up by observation, experiment and reasoning. Its main branches are the physical sciences (Physics, Chemistry), the biological sciences (Botany, Zoology) and the earth sciences; Physics itself divides into classical branches such as Mechanics, Heat, Optics and Sound, and modern branches such as Atomic, Nuclear and Plasma Physics.

(ii) Define Physics.

Physics is the branch of science that deals with the study of matter, energy and their mutual interactions. It explains how and why things move, and the laws that govern heat, light, sound, electricity and magnetism.

(iii) Name any four branches of classical Physics.

Four branches of classical Physics are: (1) Mechanics — the study of motion and forces; (2) Heat (Thermodynamics) — the study of heat energy and its conversion; (3) Optics — the study of light; (4) Sound (Acoustics) — the study of sound waves.

(iv) What is meant by Optics?

Optics is the branch of Physics that deals with the study of light, its properties, its behaviour on reflection and refraction, and the instruments based on it, such as lenses, mirrors and microscopes.

(v) Differentiate between Atomic Physics and Nuclear Physics.

Atomic Physics deals with the study of the structure and properties of atoms as a whole, especially the behaviour of electrons around the nucleus. Nuclear Physics deals with the nucleus itself — its structure, radioactivity and the energy released in nuclear reactions.

(vi) Explain briefly Solid State Physics.

Solid State Physics is the branch of Physics that deals with the study of the physical properties of solids, especially crystals and semiconductors. It explains how the

arrangement of atoms in a solid gives it electrical, magnetic and thermal properties, and it underlies devices such as transistors and computer chips.

(vii) Define the term Medical Physics.

Medical Physics is the application of the principles of Physics in medicine. It covers medical imaging (X-rays, ultrasound, MRI), radiotherapy for treating disease, and the safe use of radiation and laser instruments in diagnosis and treatment.

(viii) Differentiate between Biophysics and Geophysics.

Biophysics applies the laws of Physics to living organisms — for example, how the eye forms images, how nerves carry signals and how blood flows. Geophysics applies Physics to the Earth — the study of its internal structure, earthquakes, magnetism and the movement of its plates.

(ix) What is meant by Relativistic Mechanics?

Relativistic Mechanics is the branch of Mechanics that deals with bodies moving at speeds comparable to the speed of light. At such speeds Newton's laws need correction, as given by Einstein's theory of relativity: mass, length and time all change with speed.

(x) What is the importance of Physics in our daily life?

Physics is at work in almost everything we use daily: electricity lighting our homes, motors and engines in transport, mobile phones and the internet in communication, medical machines in hospitals, and household appliances such as refrigerators, fans and washing machines. Without Physics, modern life would not be possible.

(xi) Define the term hypothesis.

A hypothesis is a tentative, testable explanation of an observed phenomenon. It is a statement put forward to be tested by experiments; if repeated tests support it, it may grow into a theory, and if tests contradict it, it is rejected or changed.

(xii) Differentiate between theory and law.

A theory is a well-tested explanation of a natural phenomenon: when a hypothesis is tested many times with positive results, it is given the name of theory, and a theory is more certain than a hypothesis. A law is a relation among different parameters of a phenomenon which has been found empirically true and applicable in all cases — it states what happens, in a precise form.

(xiii) Explain briefly the concept of falsifiability or refutability of a theory.

Falsifiability is the capacity of a statement, assumption, hypothesis or theory to be proven wrong. A scientific theory must make predictions that can be tested: if an experiment contradicts it, the theory must be given up or corrected. A claim that cannot be tested or disproved in any way is not scientific.

(xiv) Differentiate among science, technology and engineering.

Science deals with the study of natural phenomena and the laws governing them. Technology deals with the practical application of scientific knowledge for various purposes,

such as making machines and devices. In engineering, scientific knowledge is applied to design and create practical solutions to real-world problems, such as bridges, buildings and engines.

Unit 2: Physical Quantities

Questions 2

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) The number of base units in SI is:

(a) 3

(b) 4

(c) 6

(d) 7 ✓

(ii) The SI unit of length is:

(a) metre ✓

(b) kilometre

(c) foot

(d) yard

(iii) The fundamental unit of time in SI is:

(a) day

(b) hour

(c) minute

(d) second ✓

(iv) Identify the base quantity from the following.

(a) Distance ✓

(b) Area

(c) Speed

(d) Force

(v) Which one of the followings is not a derived unit?

(a) Newton

(b) Watt

(c) Metre ✓

(d) Pascal

(vi) One cubic metre is equal to:

(a) 1000 dm^3 ✓

(b) 100 dm^3

(c) 10 dm^3

(d) 1 dm^3

(vii) One nanometre is equal to:

(a) 10^-3 m

(b) 10^-6 m

(c) 10^-9 m ✓

(d) 10^-12 m

(viii) 5 x 10^-6 s is equal to:

(a) 5 deciseconds

(b) 5 milliseconds

(c) 5 microseconds ✓

(d) 5 nanoseconds

(ix) One kilogram is equal to:

(a) 10^2 g

(b) 10^3 g ✓

(c) 10^4 g

(d) 10^5 g

(x) The mass of a dust particle is 2.5 x 10^-9 kg. How many such particles will have their combined mass equal to 2.5 kg?

(a) 10^3

(b) 10^6

(c) 10^9 ✓

(d) 10^12

(xi) In scientific notation 58000 m can be written as:

(a) 580 x 10^2 m

(b) 58 x 10^3 m

(c) 5.8 x 10^4 m ✓

(d) 5.8 x 10^-4 m

(xii) 300000000 m s^-1 in standard form is written as;

(a) 3 x 10^6 m s^-1

(b) 3 x 10^7 m s^-1

(c) 3 x 10^8 m s^-1 ✓

(d) 3 x 10^9 m s^-1

(xiii) A scalar is a physical quantity which is completely specified by

(a) number

(b) direction only

(c) number with direction

(d) number with proper unit ✓

(xiv) Which one of the followings is not a scalar quantity?

(a) distance

(b) displacement ✓

(c) speed

(d) temperature

(xv) Which one of the following is not a vector quantity?

(a) displacement

(b) speed ✓

(c) velocity

(d) acceleration

(xvi) The least count of a Vernier callipers is equal to:

(a) 1 cm

(b) 0.1 cm

(c) 0.01 cm ✓

(d) 0.001 cm

(xvii) A student claimed the diameter of a wire as 1.032 cm using Vernier callipers. Up to what extent do you agree with it?

(a) 1 cm

(b) 1.0 cm

(c) 1.03 cm ✓

(d) 1.032 cm

(xviii) The least count of a screw gauge is equal to

(a) 1 cm

(b) 0.1 cm

(c) 0.01 cm

(d) 0.001 cm ✓

(xix) Which instrument is most suitable to measure the diameter of a thin metallic wire?

(a) Screw gauge ✓

(b) Vernier callipers

(c) Measuring tape

(d) Metre rule

(xx) A mechanical stop watch can measure a time interval up to a minimum:

(a) 1 s

(b) 0.1 s ✓

(c) 0.05 s

(d) 0.01 s

(xxi) A digital stop watch can measure a time interval up to a minimum:

(a) 0.1 s ✓

(b) 0.2 s

(c) 10 s

(d) 1 s

(xxii) The number of significant figures in 23.4506 are:

(a) 2

(b) 4

(c) 5

(d) 6 ✓

(xxiii) The number of significant figures in 0.0006032 is:

(a) 7

(b) 5

(c) 4 ✓

(d) 3

B. Give brief answers to the following questions.

(i) Differentiate between physical and non-physical quantities. Give examples.

A physical quantity is one that can be measured and expressed in terms of standard units, for example length, mass, time and temperature. A non-physical quantity cannot be measured and cannot be expressed in standard units, for example love, hate, beauty and kindness.

(ii) Differentiate between base and derived quantities. Give examples.

Base quantities are those in terms of which other physical quantities can be measured; they are independent of each other — for example length, mass and time. Derived quantities are those that can be derived from base quantities by multiplication or division of two or more base quantities — for example area, volume, speed and force.

(iii) Name the seven base quantities and their SI units.

The seven base quantities and their SI units are: (1) length — metre (m); (2) mass — kilogram (kg); (3) time — second (s); (4) temperature — kelvin (K); (5) electric current — ampere (A); (6) amount of substance — mole (mol); (7) luminous intensity — candela (cd).

(iv) What is meant by derived quantities? Give examples.

Derived quantities are physical quantities that are obtained from base quantities by combining them, for example by multiplication or division. Examples are area (length x length), volume (length x length x length), speed (distance / time), density (mass / volume) and force (mass x acceleration).

(v) What is the importance of standard unit?

A standard unit makes measurements uniform, comparable and meaningful everywhere. Without agreed standard units, a measurement made in one place could not be understood or checked in another; trade, science and engineering all depend on everyone using the same units.

(vi) What is meant by scientific notation?

In scientific notation a number is expressed as a number between 1 and 10 multiplied by some power of ten. For example, 58000 m = 5.8 x 10^4 m, and 0.000000102 s = 1.02 x 10^-7 s. It is a short way of writing very large or very small numbers.

(vii) Define the term scalar quantity. Give example.

A scalar quantity is a physical quantity that is completely specified by its magnitude (a number with a proper unit) only; it has no direction. Examples are distance, speed, mass, time, temperature and energy.

(viii) Define the term vector quantity. Give examples.

A vector quantity is a physical quantity that is completely specified by its magnitude as well as its direction. Examples are displacement, velocity, acceleration, force and momentum.

(ix) What is meant by head-to-tail rule for addition of vectors?

In the head-to-tail rule, vectors are added by placing the tail of the second vector at the head (arrow end) of the first vector. The resultant (sum) vector is then drawn from the tail of the first vector to the head of the last vector; it has the same effect as all the given vectors together.

(x) Describe the head-to-tail rule for addition of three vectors.

Draw the first vector, then place the tail of the second vector at the head of the first, and the tail of the third vector at the head of the second, keeping each vector's length (magnitude) and direction unchanged. The resultant vector is drawn from the tail of the first vector to the head of the third vector.

(xi) State the least count of a Vernier callipers in mm and cm.

The least count of a Vernier callipers is 0.1 mm, which is equal to 0.01 cm. It is the smallest length the instrument can measure, and every Vernier reading is recorded correct to this least count.

(xii) State the least count of a screw gauge in mm and cm.

The least count of a screw gauge (micrometer) is 0.01 mm, which is equal to 0.001 cm. It is ten times finer than a Vernier callipers, which is why a screw gauge is used for very thin objects such as wires.

(xiii) What is meant by zero error of an instrument like Vernier callipers or screw gauge?

Zero error is the reading an instrument shows when it should read zero — for example, when the jaws of Vernier callipers are fully closed or the screw gauge spindle touches the anvil. If the zero reading is not zero, that amount must be subtracted from (or added to) every measurement to get the correct value.

(xiv) Describe how a measuring cylinder can be used to measure the volume of a solid?

Pour water into a graduated measuring cylinder and note its initial level. Tie the solid with a thread and lower it fully into the water; note the new, higher level. The rise in the water level equals the volume of the solid, because the solid displaces its own volume of water. (Volume of solid = final reading - initial reading.)

(xv) Differentiate between systematic and random errors. Give examples.

A systematic error repeats in a measurement with the same magnitude and the same sign, for example the zero error of an instrument or a wrongly calibrated scale; it can be found and corrected. A random error repeats with different magnitude and sign under the same experimental conditions, for example slight changes in the observer's eye position (parallax) or small vibrations; it is reduced by taking several readings and averaging them.

(xvi) What are significant figures?

Significant figures are the accurately known digits plus the first uncertain (doubtful) digit in a measurement. The number of significant figures is the number of digits whose values are known with certainty, together with that one doubtful digit — for example, 23.4506 has six significant figures and 0.0006032 has four.

C. Numerical Problems (every step shown).

1. A boy in your class is 1.6 m tall. What will be his height in mm and micrometre?

Step 1: 1 m = 1000 mm, so height in mm = 1.6 x 1000 = 1600 mm. Step 2: 1 mm = 1000 micrometre, so height in micrometre = 1600 x 1000 = 1,600,000 micrometre. Answer: 1600 mm and 1,600,000 micrometre.

2. The mass of neutron is 1.675 x 10^-27 kg. Convert it into gram and milligram.

Step 1: 1 kg = 1000 g = 10^3 g, so mass in gram = 1.675 x 10^-27 x 10^3 = 1.675 x 10^-24 g. Step 2: 1 g = 1000 mg = 10^3 mg, so mass in milligram = 1.675 x 10^-24 x 10^3 = 1.675 x 10^-21 mg.

3. The age of a girl studying in 9th class is 15 years. Convert her age into seconds and express this in scientific notation.

Step 1: 1 year = 365 days, so 15 years = 15 x 365 = 5475 days. Step 2: 5475 days x 24 = 131,400 hours. Step 3: 131,400 x 3600 = 473,040,000 s. Step 4: in scientific notation, 473,040,000 = 4.7304 x 10^8 s.

4. Convert one leap year into seconds and express in scientific notation.

Step 1: one leap year = 366 days. Step 2: 366 x 24 = 8784 hours. Step 3: 8784 x 3600 = 31,622,400 s. Step 4: in scientific notation, 31,622,400 = 3.16224 x 10^7 s.

5. Express the following quantities in scientific notation. (a) 5 000 000 000 000 m (b) 21 033 000 000 kg (c) 0.000 000 000 246 m (d) 0.000 000 102 s

Move the decimal point until one non-zero digit is left before it, and count the places moved. (a) 5,000,000,000,000 m = 5 x 10^12 m. (b) 21,033,000,000 kg = 2.1033 x 10^10 kg. (c) 0.000000000246 m = 2.46 x 10^-10 m. (d) 0.000000102 s = 1.02 x 10^-7 s.

6. Express the following quantities as ordinary whole number. (a) 2.34 x 10^5 m (b) 1.8 x 10^-6 kg (c) 2.5 x 10^-9 s

(a) 2.34 x 10^5 m = 234000 m. (b) 1.8 x 10^-6 kg = 0.0000018 kg. (c) 2.5 x 10^-9 s = 0.0000000025 s. (Note: the book mislabels the third part as (d); it is answered here as (c).)

7. Draw the representative lines of the following vector quantities on a suitable

scale. (a) Displacement 100 m, 60 degrees with x-axis (b) Velocity 30 m s^-1, North (c) Acceleration 2.5 m s^-2, 135 degrees with x-axis (d) Force 200 N, North East

Choose a scale, for example 1 cm = 20 m for displacement, 1 cm = 10 m s^-1 for velocity, 1 cm = 0.5 m s^-2 for acceleration and 1 cm = 50 N for force. Then draw each arrow with length = magnitude / scale, pointing in the given direction: (a) 5 cm at 60 degrees above the x-axis; (b) 3 cm pointing North; (c) 5 cm at 135 degrees from the x-axis; (d) 4 cm pointing North-East. Each arrow's length shows its magnitude and its direction shows the vector's direction.

8. Two forces of magnitude 12 N and 5 N are acting on a body along x-axis and y-axis respectively. Find the magnitude and the direction of the resultant force graphically or by calculation.

Step 1: the two forces are perpendicular, so the resultant R = sqrt(Fx^2 + Fy^2) = sqrt(12^2 + 5^2) = sqrt(144 + 25) = sqrt(169) = 13 N. Step 2: direction: tan(theta) = Fy / Fx = 5 / 12 = 0.4167, so theta = 22.6 degrees with the x-axis. Answer: 13 N at 22.6 degrees to the x-axis.

9. What is the reading on Vernier callipers for the following figure?

From the figure, the main-scale reading is 4.3 cm and the Vernier division that coincides with a main-scale division adds 0.05 cm. Step: reading = main scale + Vernier coincidence x least count = 4.3 + 0.05 = 4.35 cm. Answer: 4.35 cm.

10. What is the pitch of the micrometer shown in the figure. Find the reading on screw gauge for the following figure?

From the figure the screw-gauge reading is worked out from its main scale and the circular scale division that coincides with the reference line, using the instrument's least count of 0.01 mm. The reading so obtained is 5.78 mm, as printed with the question. Answer: 5.78 mm.

11. State the number of significant figures in the following measurements. (a) 234567 m (b) 2003 kg (c) 0.000529 s (d) 5.98 x 10^24 kg (e) 15.30 J

(a) 234567: all six digits are significant = 6. (b) 2003: the zeros between non-zero digits count = 4. (c) 0.000529: leading zeros do not count; 5, 2, 9 count = 3. (d) 5.98 x 10^24: the coefficient 5.98 has 3 significant figures (the power of ten does not add any). (e) 15.30: the trailing zero after the decimal point counts = 4. Answers: (a) 6, (b) 4, (c) 3, (d) 3, (e) 4. (Note: the book's printed answer line for this question is garbled — it lists only four entries and repeats (c); the count above follows the significant-figure rules.)

12. The radius of a metallic sphere is 2.35 cm. Calculate its volume using the formula V = 4/3 pi r^3 up to reasonable number of significant numbers.

Step 1: V = (4/3) x pi x r^3 = (4/3) x 3.14 x (2.35)^3. Step 2: (2.35)^3 = 12.977. Step 3: V = 4.1888 x 12.977 = 54.36 cm^3. Step 4: the radius has three significant figures, so round

to three significant figures: 54.4 cm^3. Answer: 54.4 cm^3.

13. The length, width and height of a rectangular wooden block is 5.50 cm, 2.25 cm and 5.1 cm respectively. Calculate the volume of block correct upto appropriate number of significant figures.

Step 1: V = length x width x height = 5.50 x 2.25 x 5.1. Step 2: 5.50 x 2.25 = 12.375. Step 3: 12.375 x 5.1 = 63.11 cm^3. Step 4: the least precise measurement (5.1 cm) has two significant figures, so round to two significant figures: 63 cm^3. Answer: 63 cm^3.

14. Round off the following numbers to three significant figures. (a) 8.724 (b) 8.725 (c) 8.735 (d) 8.737

Keep three significant figures and round on the fourth digit. (a) 8.724 -> 8.72 (fourth digit 4, round down). (b) 8.725 -> 8.72 (as printed in the book). (c) 8.735 -> 8.74 (fourth digit 5, round up). (d) 8.737 -> 8.74 (fourth digit 7, round up). Answers: (a) 8.72, (b) 8.72, (c) 8.74, (d) 8.74.

Unit 3: Kinematics

Questions 3

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) A body has translatory motion if it moves along a:

(a) straight line

(b) circle

(c) line without rotation ✓

(d) curved path

(ii) One of the followings is an example of random motion.

(a) Motion of a car

(b) Motion of pendulum

(c) Flight of a butterfly ✓

(d) Motion of rocket

(iii) The periodic motion of the pendulum of a clock is an example of

(a) rotatory motion

(b) translatory motion

(c) vibratory motion ✓

(d) linear motion

(iv) The spinning of Earth is an example of:

(a) rotatory motion ✓

(b) translatory motion

(c) circular motion

(d) vibratory motion

(v) The motion of blades of electric fan is an example of:

(a) Translatory motion

(b) Vibratory motion

(c) Linear motion

(d) Rotatory motion ✓

(vi) The shortest distance from initial point to final point is known as:

(a) position

(b) distance

(c) displacement ✓

(d) speed

(vii) The distance covered by a body in unit time is called;

(a) acceleration

(b) velocity

(c) average velocity

(d) speed ✓

(viii) The time rate of change of displacement is termed as:

(a) acceleration

(b) velocity ✓

(c) speed

(d) total distance covered

(ix) A train is moving at a speed of 36 km h^-1. Its speed expressed in m s^-1 is:

(a) 10 m s^-1 ✓

(b) 20 m s^-1

(c) 25 m s^-1

(d) 30 m s^-1

(x) If a body covers equal distance in equal intervals of time, however small the intervals may be, the speed of body is said to be:

(a) uniform ✓

(b) variable

(c) average

(d) non-uniform

(xi) The time rate of change of velocity of body is known as;

(a) acceleration ✓

(b) average velocity

(c) displacement

(d) total distance covered

(xii) A car travelling at 10 m s^-1 accelerates uniformly at 2 m s^-2. What will be its velocity after 5 s?

(a) 2 m s^-1

(b) 15 m s^-1

(c) 20 m s^-1 ✓

(d) 40 m s^-1

(xiii) The speed of a car increases from 20 m s^-1 to 50 m s^-1 in 10 seconds. Its acceleration

(a) 2 m s^-2

(b) 3 m s^-2 ✓

(c) 5 m s^-2

(d) 7 m s^-2

(xiv) The distance-time graph for an object moving with variable speed is a

(a) straight line parallel to vertical axis

(b) straight line parallel to time axis

(c) straight line passing through origin

(d) curved line ✓

(xv) If an object is moving with uniformly changing speed, then the slope of speed-time graph determines the magnitude of;

(a) distance

(b) displacement

(c) velocity

(d) acceleration ✓

(xvi) Which of the following graph represents uniform acceleration: (a) distance-time straight line through origin, (b) distance-time horizontal line, (c) velocity-time straight line through origin, (d) velocity-time curve levelling off

(a) graph (a)

(b) graph (b)

(c) graph (c) ✓

(d) graph (d)

(xvii) A cyclist starting from rest moves with an acceleration of 2 m s^-2. What distance will it cover in 10 seconds?

(a) 200 m

(b) 100 m ✓

(c) 20 m

(d) 10 m

(xviii) Two balls of different masses are dropped freely from a tower. Which ball will reach the ground first?

(a) light ball

(b) heavy ball

(c) both balls will reach at the same time ✓

(d) heavy ball will reach the ground first

(xix) A ball was dropped from a tower freely. The distance covered in first second is;

(a) 4.9 m ✓

(b) 9.8 m

(c) 49 m

(d) 98 m

(xx) A ball is thrown vertically upward. Its velocity at the highest point is

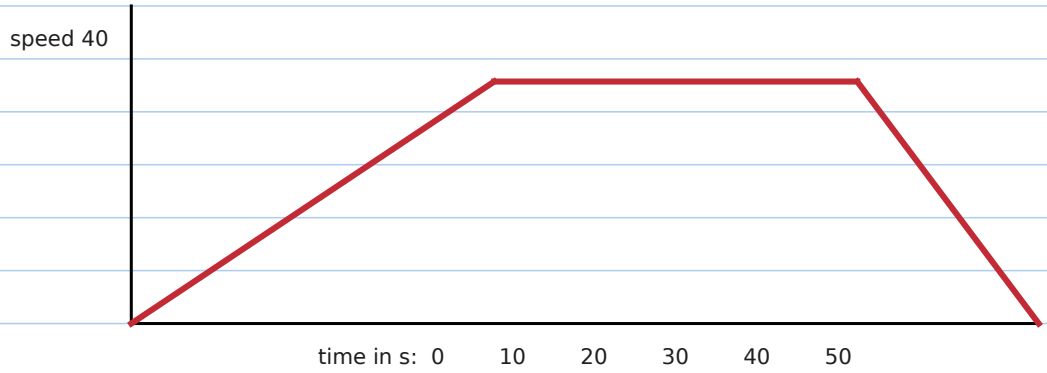
(a) -9.8 m s^-1

(b) zero ✓

(c) 9.8 m s^-1

(d) None of these

Diagram: speed-time graph for Unit 3, Numerical Problem 13



B. Give brief answers to the following questions.

(i) Define the term kinematics.

Kinematics is the branch of Mechanics that deals with the study of motion of bodies without reference to the forces that cause the motion. It describes motion in terms of displacement, velocity, acceleration and time.

(ii) Differentiate between rest and motion. Give examples.

A body is said to be at rest if it does not change its position with respect to its surroundings with time — for example, a book lying on a table. A body is said to be in motion if it changes its position with respect to its surroundings with time — for example, a moving car or a flying bird.

(iii) How many types of motion are there?

There are three main types of motion: (1) translatory motion, in which a body moves along a line without rotation; (2) rotatory motion, in which a body spins about a fixed axis; (3) vibratory motion, in which a body moves to and fro about a mean position. Random motion is also commonly listed as a further type.

(iv) Describe three types of translatory motion.

The three types of translatory motion are: (1) linear motion — motion along a straight line, for example a car on a straight road; (2) circular motion — motion along a circular path, for example a stone tied to a string whirled in a circle; (3) random motion — motion with no fixed path or direction, for example the flight of a butterfly or a bee.

(v) Define the term rotatory motion with suitable examples.

Rotatory motion is the motion in which a body spins or rotates about a fixed axis, while its particles move in circles around that axis. Examples are the spinning of the Earth on its axis, the blades of an electric fan, and a spinning wheel.

(vi) Define vibratory motion and give its examples from daily life.

Vibratory motion is the to-and-fro motion of a body about its mean (rest) position, repeated after equal intervals of time. Examples from daily life are the pendulum of a clock, a vibrating guitar string, and a swing.

(vii) Differentiate between distance and displacement.

Distance is the total length of the path actually travelled by a body; it is a scalar quantity and is always positive. Displacement is the shortest distance from the initial point to the final point, in a stated direction; it is a vector quantity and can be positive, negative or zero.

(viii) Differentiate between speed and velocity.

Speed is the distance covered by a body in unit time; it is a scalar quantity and has no direction. Velocity is the time rate of change of displacement, that is, displacement per unit time in a stated direction; it is a vector quantity.

(ix) Define the term acceleration.

Acceleration is the time rate of change of velocity of a body. It is a vector quantity, measured in metre per second squared (m s^-2), and it is positive when velocity increases and negative (retardation) when velocity decreases.

(x) The distance-time graph of an object is a horizontal line parallel to time axis. What do you conclude from it?

A horizontal line parallel to the time axis means the distance does not change as time passes, so the object is at rest: its speed is zero throughout.

(xi) The distance-time graph of a moving object is an inclined straight line with time axis. How will you calculate the speed of the moving object?

For an inclined straight distance-time graph, the speed is the slope (gradient) of the line: speed = change in distance / change in time, read off any two points on the line.

(xii) What will be the shape of speed-time graph of an object which is moving with constant acceleration?

For constant acceleration, the speed increases uniformly with time, so the speed-time graph is a straight line inclined to the time axis, passing through the origin if the object starts from rest.

(xiii) What is represented by the area under a speed-time graph of a moving object?

The area under a speed-time graph represents the distance covered by the moving object in that time interval. For a velocity-time graph it represents the displacement.

(xiv) Derive first equation of motion using speed-time graph.

For motion with uniform acceleration a, the speed-time graph is a straight line of slope a. Slope = change in speed / time = (vf - vi) / t, so a = (vf - vi) / t, which gives vf = vi + at — the first equation of motion.

(xv) Derive second equation of motion using speed-time graph.

The distance covered equals the area under the speed-time graph, which is a trapezium of area = (vi + vf) / 2 x t. Putting vf = vi + at gives S = vi t + (1/2) a t^2 — the second equation of motion.

(xvi) Derive third equation of motion using speed-time graph.

From vf = vi + at we get t = (vf - vi) / a. Putting this in S = (vi + vf) / 2 x t gives S = (vf^2 -

vi^2) / (2a), so 2aS = vf^2 - vi^2 — the third equation of motion.

(xvii) What is meant by acceleration due to gravity? State its value.

Acceleration due to gravity is the uniform acceleration of a freely falling body under the Earth's gravitational pull, when air resistance is ignored. Its value near the Earth's surface is g = 9.8 m s^-2, directed towards the centre of the Earth.

(xviii) What is the universal speed limit for any object in the universe?

The universal speed limit is the speed of light in vacuum, c = 3 x 10^8 m s^-1. According to relativity, no material object can reach or exceed this speed.

C. Numerical Problems (every step shown).

1. A girl travels a distance of 20 m towards West, then 30 m towards North and finally 20 m towards East. Calculate; (a) total distance travelled. (b) the resultant displacement.

Step 1: total distance = 20 + 30 + 20 = 70 m. Step 2: the 20 m West and 20 m East cancel each other, leaving only the 30 m North. Answer: (a) 70 m, (b) 30 m towards North.

2. A car travels 100 metres in 5 seconds. What is its average speed?

Step 1: average speed = total distance / total time. Step 2: = 100 / 5 = 20 m s^-1. Answer: 20 m s^-1.

3. Express the speed 90 km h^-1 into m s^-1.

Step 1: 1 km = 1000 m and 1 h = 3600 s, so divide by 3.6. Step 2: 90 / 3.6 = 25 m s^-1. Answer: 25 m s^-1.

4. Express the speed 30 m s^-1 into km h^-1.

Step 1: multiply by 3.6 to go from m s^-1 to km h^-1. Step 2: 30 x 3.6 = 108 km h^-1. Answer: 108 km h^-1.

5. A car accelerates from 5 m s^-1 to 25 m s^-1 in 10 seconds. What is its acceleration?

Step 1: a = (vf - vi) / t. Step 2: a = (25 - 5) / 10 = 20 / 10 = 2 m s^-2. Answer: 2 m s^-2.

6. A car moving with velocity of 72 km h^-1 is brought to rest in 10 s. Calculate the retardation of the car.

Step 1: convert: 72 km h^-1 = 72 / 3.6 = 20 m s^-1. Step 2: a = (0 - 20) / 10 = -2 m s^-2. The negative sign shows retardation. Answer: -2 m s^-2 (retardation 2 m s^-2).

7. A car starts up from rest. If acceleration of car is 5 m s^-2 then what will be its velocity after 12 s.

Step 1: vf = vi + at, with vi = 0. Step 2: vf = 0 + 5 x 12 = 60 m s^-1. Answer: 60 m s^-1.

8. The velocity of a motor cycle becomes 20 m s^-1 in 4 s with an acceleration of 2 m s^-2. Find its initial velocity.

Step 1: vf = vi + at, so vi = vf - at. Step 2: vi = 20 - 2 x 4 = 20 - 8 = 12 m s^-1. Answer: 12 m s^-1.

9. A driver brings a car travelling at 20 m s^-1 to stop in 2.0 seconds. (a) What is

the acceleration of car? (b) How far does car travel before stopping?

Step 1: a = (0 - 20) / 2 = -10 m s^-2. Step 2: use 2aS = vf^2 - vi^2, so S = (0 - 400) / (2 x -10) = 20 m. Answer: (a) -10 m s^-2, (b) 20 m.

10. A racing car is moving with 180 km h^-1. Brakes are applied and a negative acceleration of 10 m s^-2 is produced. Calculate the distance covered by the car before coming to rest.

Step 1: convert: 180 km h^-1 = 180 / 3.6 = 50 m s^-1. Step 2: S = (0 - 50^2) / (2 x -10) = (-2500) / (-20) = 125 m. Answer: 125 m.

11. A body starts from rest and acquires a velocity of 8 m s^-1 in 5 s. Calculate (a) the acceleration (b) the distance covered in 5 s (c) the distance covered in 5th second.

Step 1: a = (8 - 0) / 5 = 1.6 m s^-2. Step 2: S in 5 s = (1/2) a t^2 = 0.5 x 1.6 x 25 = 20 m. Step 3: distance in 4 s = 0.5 x 1.6 x 16 = 12.8 m, so distance in the 5th second = 20 - 12.8 = 7.2 m. Answers: (a) 1.6 m s^-2, (b) 20 m, (c) 7.2 m.

12. An airplane must acquire a velocity of 71 m s^-1 for take-off. If the runway is 1.0 km long, what must be the acceleration of airplane.

Step 1: use 2aS = vf^2 - vi^2 with vi = 0, S = 1000 m. Step 2: a = 71^2 / (2 x 1000) = 5041 / 2000 = 2.52 m s^-2. Answer: 2.52 m s^-2.

13. The figure shows a speed-time graph for a motor cycle. Find; (a) What is the maximum speed of the motorcycle. (b) For how many seconds does the motorcycle stay at its maximum speed? (c) For how many seconds is the motorcycle actually moving? (d) How much speed does the motorcycle gain in first 20 seconds? How much speed does it gain every second? What is its acceleration? (e) What is the retardation of the motorcycle during the last 10 seconds.

Read the graph: speed rises from 0 to 40 m s^-1 in the first 20 s, stays at 40 m s^-1 from 20 s to 40 s, then falls to 0 in the last 10 s (40 s to 50 s). (a) Maximum speed = 40 m s^-1. (b) At maximum speed for 40 - 20 = 20 s. (c) Moving over the whole 50 s shown. (d) Speed gained in first 20 s = 40 m s^-1, so 40 / 20 = 2 m s^-1 every second; acceleration = 2 m s^-2. (e) Retardation in the last 10 s = (0 - 40) / 10 = -4 m s^-2. (A vector sketch of this speed-time graph is drawn with Unit 3 in the notes.)

14. The table gives speed and time for a train as it travels from one station to the next. Time (s): 0, 20, 40, 60, 80, 100, 120, 140, 160. Speed (m s^-1): 0, 5, 10, 15, 20, 20, 20, 10, 0. (a) Plot a speed-time graph using above data. (b) Calculate the acceleration in each of the three parts of the journey. (c) Calculate the total distance travelled by the train.

(a) Plot time on the x-axis and speed on the y-axis and join the points: the graph rises steadily to 20 m s^-1 at 80 s, stays flat until 120 s, then falls to 0 at 160 s. (b) First part (0

to 80 s): a = (20 - 0) / 80 = 0.25 m s^-2. Second part (80 to 120 s): speed constant, so a = 0. Third part (120 to 160 s): a = (0 - 20) / 40 = -0.5 m s^-2. (c) Distance = area under the graph = triangle (0.5 x 80 x 20 = 800 m) + rectangle (40 x 20 = 800 m) + triangle (0.5 x 40 x 20 = 400 m) = 2000 m.

15. A stone is dropped from height of 80 m. Find its velocity just before hitting the ground if it reaches the ground in 5 seconds. (Take the value g = 9.8 m s^-2)

Step 1: vf = vi + gt, with vi = 0. Step 2: vf = 0 + 9.8 x 5 = 49 m s^-1. Answer: 49 m s^-1. (Note: the stated 5 s and 80 m are inconsistent with each other — 80 m of free fall takes about 4.0 s — but the answer follows the question's stated time, as printed.)

16. A ball is thrown up vertically with a speed of 49 m s^-1. (a) How long does it take to reach its highest point? (b) How high does the ball rise? (Take the value g = 9.8 m s^-2)

Step 1: at the highest point vf = 0, so t = (vf - vi) / (-g) = (0 - 49) / (-9.8) = 5 s. Step 2: h = vi^2 / (2g) = 49^2 / (2 x 9.8) = 2401 / 19.6 = 122.5 m. Answers: (a) 5 s, (b) 122.5 m.

Unit 4: Dynamics and Forces

Questions 4

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) The part of mechanics that deals with the study of motion with the reference of mass and force is called;

(a) kinematics

(b) dynamics ✓

(c) thermodynamics

(d) quantum mechanics

(ii) What one of the followings is the measure of inertia?

(a) weight

(b) force

(c) velocity

(d) mass ✓

(iii) Which of the followings remains constant everywhere?

(a) weight

(b) pressure

(c) mass ✓

(d) velocity

(iv) Which of the following instruments is used to measure mass?

(a) physical balance ✓

(b) spring balance

(c) Vernier callipers

(d) measuring cylinder

(v) The property of matter by which it resists any change in its state of rest or uniform motion is called;

(a) velocity

(b) force

(c) momentum

(d) inertia ✓

(vi) Which of the followings is not constant everywhere?

(a) weight ✓

(b) pressure

(c) mass

(d) velocity

(vii) Which of the following instruments is used to measure weight?

(a) physical balance

(b) spring balance ✓

(c) Vernier callipers

(d) measuring cylinder

(viii) The mass of an object is 10 kg. What will be its weight on Earth?

(a) 4.9 N

(b) 9.8 N ✓

(c) 15 N

(d) 20 N

(ix) Which of the following law is also called law of inertia?

(a) Newton's first law of motion ✓

(b) Newton's second law of motion

(c) Newton's third law of motion

(d) Newton's law of gravitation

(x) A man jumps from a moving bus. He will fall;

(a) towards the moving bus

(b) away from the bus

(c) in the direction of motion ✓

(d) opposite to the direction of motion

(xi) Which of the followings is true according to Newton's second law of motion?

(a) m = a / F

(b) F = a / m

(c) F = m / a

(d) m = F / a ✓

(xii) If mass of a body is doubled while the force is kept constant, then the acceleration will be;

(a) one half ✓

(b) doubled

(c) one fourth

(d) four times

(xiii) When a force of 20 N acts on a mass of 4 kg the acceleration will be;

(a) 2 m s^-2

(b) 5 m s^-2 ✓

(c) 24 m s^-2

(d) 80 m s^-2

(xiv) The SI unit of force is

(a) watt

(b) joule

(c) newton ✓

(d) pascal

(xv) Action and reaction are equal but

(a) in same direction

(b) in opposite directions ✓

(c) perpendicular to each other

(d) none of these

(xvi) Motion of a rocket is governed by;

(a) Newton's first law of motion

(b) Newton's second law of motion

(c) Newton's third law of motion ✓

(d) Newton's law of gravitation

(xvii) The force which resists the motion of one surface on another surface is known as;

(a) Momentum

(b) weight

(c) mass

(d) friction ✓

(xviii) The resisting force between the two surfaces before the motion starts is called:

(a) Kinetic friction

(b) Rolling friction

(c) Static friction ✓

(d) sliding friction

(xix) The rolling friction is much _____ then the sliding friction.

(a) less ✓

(b) greater

(c) equal

(d) equal or greater

(xx) Which type of friction has a minimum value?

(a) Static friction

(b) Kinetic friction

(c) Rolling friction ✓

(d) Limiting friction

(xxi) What is the purpose of using lubricant between moving parts of a machine?

(a) To decrease the friction ✓

(b) To make the friction zero

(c) To increase the friction

(d) none of these

B. Give brief answers to the following questions.

(i) What is meant by mass of an object?

Mass is the quantity of matter in an object. It is a scalar quantity, measured in kilograms, it does not change from place to place, and it is the measure of the object's inertia — the greater the mass, the harder it is to change the object's state of motion.

(ii) Define the term inertia.

Inertia is the property of matter by which a body resists any change in its state of rest or of uniform motion in a straight line. A body stays at rest or keeps moving uniformly unless an unbalanced external force acts on it.

(iii) Give at least two examples of inertia from daily life.

Examples of inertia: (1) when a bus suddenly starts, passengers fall backwards because their bodies tend to stay at rest; (2) when a moving bus suddenly stops, passengers fall forwards because their bodies tend to keep moving; (3) dust is removed from a carpet by beating it, because the dust tends to stay at rest while the carpet moves.

(iv) Which instrument is used to measure the mass of an object?

The mass of an object is measured with a physical balance (beam balance), which compares the unknown mass with standard masses. A spring balance measures weight, not mass.

(v) What happens when a force acts on an object?

When a force acts on an object it can change the object's state of rest or of uniform motion, change its speed or direction, change its shape or size, or stop a moving object. In short, a force produces acceleration or deformation.

(vi) Define the term force.

Force is an agency that changes, or tends to change, the state of rest or of uniform motion of a body, or its shape. It is a vector quantity, measured in newtons (N); one newton is the force that gives a mass of 1 kg an acceleration of 1 m s^-2.

(vii) Differentiate between contact and non-contact forces.

Contact forces act only when the bodies are in physical contact, for example friction, normal reaction, tension and muscular force. Non-contact forces act without contact, through a field, for example gravitational force, electrostatic force and magnetic force.

(viii) Give at least two examples of contact forces.

Examples of contact forces are: (1) friction between a book and a table; (2) the normal reaction of a table on a book resting on it; (3) tension in a rope pulling a load; (4) muscular force applied when pushing a cart.

(ix) Give at least two examples of non-contact forces.

Examples of non-contact forces are: (1) the gravitational force of the Earth pulling objects downwards; (2) the magnetic force between two magnets; (3) the electrostatic force

between charged bodies.

(x) Define weight. How it is related to mass of an object?

Weight is the force with which the Earth (or another planet) attracts a body towards its centre. It is related to mass by W = mg, where m is the mass and g is the acceleration due to gravity (9.8 m s^-2 on Earth). Weight is a vector quantity measured in newtons, and it changes from place to place because g changes.

(xi) Give a comparison of mass and weight.

Mass is the quantity of matter in a body, a scalar measured in kilograms, and it is constant everywhere. Weight is the gravitational force on the body, a vector measured in newtons, given by W = mg; it changes with g, so it is less on the Moon than on Earth and zero in free space.

(xii) Define the term gravitational field. Give its SI unit.

The gravitational field is the region around a mass in which another mass experiences a gravitational force. Its strength at a point is the force per unit mass there, g = F / m. Its SI unit is newton per kilogram (N kg^-1), which is the same as m s^-2.

(xiii) Name the fundamental forces of nature.

The four fundamental forces of nature are: (1) gravitational force; (2) electromagnetic force; (3) strong nuclear force, which holds protons and neutrons together in the nucleus; (4) weak nuclear force, which is responsible for radioactive decay.

(xiv) State and explain Newton's first law of motion.

Newton's first law states: a body remains at rest or in uniform motion in a straight line unless acted upon by an unbalanced external force. This is why it is called the law of inertia — it is the body's inertia that keeps it in its state until a net force changes it.

(xv) Give at least two examples where Newton's first law of motion is valid.

Examples: (1) a book stays at rest on a table until someone pushes it; (2) a passenger falls backwards when a bus suddenly starts and forwards when it suddenly stops; (3) a hockey puck slides on smooth ice at nearly constant velocity until friction or a stick stops it.

(xvi) State and explain Newton's second law of motion.

Newton's second law states: when a net force acts on a body, the acceleration produced is directly proportional to the force and inversely proportional to the mass, and it takes place in the direction of the force: F = ma. So a larger force gives a larger acceleration, and a larger mass gives a smaller acceleration for the same force.

(xvii) Define one newton.

One newton is the force which, acting on a mass of 1 kg, produces an acceleration of 1 m s^-2 in the direction of the force. Since F = ma, 1 N = 1 kg x 1 m s^-2 = 1 kg m s^-2.

(xviii) State and explain Newton's third law of motion.

Newton's third law states: to every action there is an equal and opposite reaction. When one body exerts a force (action) on a second body, the second body exerts an equal force

(reaction) back on the first, in the opposite direction. The two forces act on different bodies, so they never cancel each other.

(xix) Give at least two examples of Newton's third law of motion.

Examples: (1) a rocket moves forward because it pushes exhaust gases backwards and the gases push the rocket forwards; (2) when you walk, your foot pushes the ground backwards and the ground pushes you forwards; (3) a gun recoils backwards when a bullet is fired forwards.

(xx) Differentiate among stable, unstable and neutral equilibriums.

In stable equilibrium a body returns to its original position after a small displacement, for example a cone resting on its base. In unstable equilibrium a small displacement moves the body further away and it does not return, for example a cone balanced on its tip. In neutral equilibrium a body stays in its new position after displacement, for example a cone lying on its side rolled slightly.

(xxi) What is meant by friction?

Friction is the force that resists the relative motion (or the tendency to move) of one surface over another surface in contact with it. It acts parallel to the surfaces and opposite to the direction of motion, and it arises from the roughness and interlocking of the surfaces.

(xxii) Differentiate between static and limiting friction.

Static friction is the friction that acts between two surfaces at rest relative to each other; it adjusts itself to balance the applied force, up to a maximum value. Limiting friction is that maximum value of static friction, just before the surfaces start to slide; once motion begins, friction becomes kinetic friction.

(xxiii) State and briefly explain sliding or kinetic friction.

Sliding (kinetic) friction is the friction that acts between two surfaces while they are sliding over each other. It is slightly less than limiting friction, it opposes the sliding motion, and its value is nearly independent of the sliding speed and of the area of contact.

(xxiv) State and briefly explain rolling friction.

Rolling friction is the friction that acts when a body rolls over a surface, for example a wheel on a road. It is much smaller than sliding friction, which is why wheels, ball bearings and rollers are used to reduce friction in machines and vehicles.

(xxv) What are advantages of friction?

Advantages of friction: (1) we can walk and run without slipping; (2) vehicles can move and brake on roads; (3) we can grip and hold objects and write with a pen; (4) nails and screws stay fixed in wood; (5) belts drive machines without slipping.

(xxvi) What are disadvantages of friction?

Disadvantages of friction: (1) it wastes energy as heat and reduces the efficiency of machines; (2) it wears out moving parts such as tyres, brake pads and engine parts; (3) it opposes motion, so extra force and fuel are needed; (4) it produces unwanted noise and

heat.

(xxvii) Give at least two methods of reducing friction.

Methods of reducing friction: (1) use lubricants such as oil or grease between moving parts; (2) use ball bearings or rollers to replace sliding with rolling; (3) polish and smooth the surfaces in contact; (4) streamline the shape of fast-moving bodies; (5) use an air cushion where possible.

C. Numerical Problems (every step shown).

1. The mass of an object is 1500 kg. What is its weight on Earth? Take the value of 'g' as 9.8 m s^-2.

Step 1: W = mg. Step 2: W = 1500 x 9.8 = 14700 N. Step 3: in scientific notation, 14700 = 1.47 x 10^4 N. Answer: 1.47 x 10^4 N.

2. The weight of an object on Earth is 980 N. Calculate its mass.

Step 1: m = W / g. Step 2: m = 980 / 9.8 = 100 kg. Answer: 100 kg. (Note: the book's printed answer of 10 kg is wrong — 980 / 9.8 = 100, not 10.)

3. A block of 100 kg is pushed horizontally by a force of 250 N. Calculate the acceleration produced in the block.

Step 1: a = F / m. Step 2: a = 250 / 100 = 2.5 m s^-2. Answer: 2.5 m s^-2.

4. A wooden block of mass 5 kg is pushed along a smooth surface of a table with a force of 30 N, find the acceleration of the block.

Step 1: a = F / m. Step 2: a = 30 / 5 = 6 m s^-2. Answer: 6 m s^-2.

5. A car of mass 1500 kg is moving with velocity of 20 m s^-1. It is stopped in 5 seconds by applying brakes. Find: (a) What is the average acceleration? (b) How much force is applied? (c) How much distance is covered?

Step 1: average acceleration a = (0 - 20) / 5 = -4 m s^-2 (magnitude 4 m s^-2). Step 2: force F = ma = 1500 x 4 = 6000 N. Step 3: distance S = average velocity x time = ((20 + 0) / 2) x 5 = 10 x 5 = 50 m. Answers: (a) 4 m s^-2, (b) 6000 N, (c) 50 m. (Note: the book's printed answer for (c) is 40 m, which does not follow from its own data — the correct distance is 50 m.)

Unit 5: Momentum

Questions 5

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) The quantity of motion in an object is called its:

(a) mass

(b) velocity

(c) acceleration

(d) momentum ✓

(ii) Momentum is equal to the product of mass and:

(a) force

(b) velocity ✓

(c) acceleration

(d) distance

(iii) The product of mass and velocity is called:

(a) force

(b) impulse

(c) momentum ✓

(d) torque

(iv) The direction of momentum is:

(a) along the velocity ✓

(b) opposite to velocity

(c) perpendicular to velocity

(d) none of these

(v) The SI units of momentum are:

(a) kg m s

(b) kg m s^-1 ✓

(c) kg m s^-2

(d) kg m^-1 s

(vi) The SI unit of momentum is equal to:

(a) N s ✓

(b) N^-1 s

(c) N s^-1

(d) kg m^-1 s

(vii) In the absence of external force, the momentum of a system before collision is:

(a) less than momentum after collision

(b) greater than momentum after collision

(c) equal to momentum after collision ✓

(d) equal to zero

(viii) The momentum of an isolated system has:

(a) minimum value

(b) maximum value

(c) zero value

(d) constant value ✓

(ix) The rate of change of momentum is equal to:

(a) mass

(b) velocity

(c) applied force ✓

(d) impulse

(x) Impulse is defined as product of force and:

(a) mass

(b) velocity

(c) time ✓

(d) acceleration

(xi) The SI unit of impulse is equal to:

(a) N s ✓

(b) N^-1 s

(c) N s^-1

(d) N^-1 s^-1

B. Give brief answers to the following questions.

(i) Define the term momentum and give its SI units.

Momentum is the quantity of motion in an object, defined as the product of its mass and its velocity: p = mv. It is a vector quantity in the direction of the velocity. Its SI unit is kilogram metre per second (kg m s^-1), which is the same as newton second (N s).

(ii) State whether momentum is scalar or vector quantity.

Momentum is a vector quantity, because velocity is a vector: momentum has both magnitude (mv) and direction, and its direction is always the direction of the velocity of the body.

(iii) A cricket ball and a tennis ball are moving with same velocity. Explain why it is difficult to stop the cricket ball.

Both balls have the same velocity, but the cricket ball has a much larger mass, so its momentum (mv) is much larger. Stopping it needs a much larger impulse (force x time), so it is harder to stop than the lighter tennis ball.

(iv) What is meant by an isolated system? Give at least one example.

An isolated system is a system on which no external (outside) force acts, so its total momentum stays constant. Examples are two colliding billiard balls on a smooth table (ignoring friction), or a bullet and gun before and after firing, taken together.

(v) State the law of conservation of momentum.

The law of conservation of momentum states: the total momentum of an isolated system remains the same (constant), provided that no external force acts upon it. In a collision, the total momentum before the collision equals the total momentum after the collision.

(vi) When a bullet is fired from a gun then gun recoils. Explain.

Before firing, the total momentum of the gun and bullet is zero. When the bullet goes forward with momentum mv, the gun must gain an equal momentum in the opposite direction so that the total stays zero: M V = -m v. That backward motion of the gun is its recoil.

(vii) State Newton's second law of motion in terms of momentum.

In terms of momentum, Newton's second law states: the force acting on a body is equal to the rate of change of its momentum, and the change takes place in the direction of the applied force: F = (mvf - mvi) / t = change in momentum / time.

(viii) What do you understand by the term impulse of a force?

When a large force F acts on an object for an extremely short interval of time (delta t), the product F x (delta t) is called the impulse of the force. Impulse equals the change in momentum produced: impulse = F x (delta t) = m x (change in velocity). Its unit is N s.

(ix) Give two examples from daily life involving impulse.

Examples of impulse: (1) a cricket player lowers his hands while catching a fast ball, increasing the time of catch so the force on his hands is smaller; (2) heavy vehicles have spring suspension at the bottom, which increases the time of a jerk and reduces the force; (3) a karate player breaks a brick with a very short, hard blow.

(x) Does a moving object have impulse?

No. A moving object has momentum, not impulse. Impulse is not something a body possesses: it is the product of a force and the short time for which it acts, and it equals the change in momentum that the force produces. Without a force acting, there is no impulse, even though the body keeps its momentum.

C. Numerical Problems (every step shown).

1. The mass of a sprinter is 45 kg. What will be the momentum of a sprinter if he run at a velocity of 36 km h^-1?

Step 1: convert velocity: 36 km h^-1 = 36 / 3.6 = 10 m s^-1. Step 2: p = mv = 45 x 10 = 450 kg m s^-1. Answer: 450 kg m s^-1.

2. An object of mass 250 kg is moving with a velocity of 10 m s^-1. A constant force acts for 10 seconds and increases its velocity to 20 m s^-1. Calculate the momentum of the object before and after application of the force.

Step 1: momentum before = 250 x 10 = 2500 = 2.5 x 10^3 kg m s^-1. Step 2: momentum after = 250 x 20 = 5000 = 5 x 10^3 kg m s^-1. Answers: 2.5 x 10^3 kg m s^-1 before, 5 x 10^3 kg m s^-1 after.

3. A car is moving on a straight flat road with a velocity 108 km h^-1. Find the mass of this car if its momentum is 4.5 x 10^4 kg m s^-1.

Step 1: convert velocity: 108 / 3.6 = 30 m s^-1. Step 2: m = p / v = 4.5 x 10^4 / 30 = 1500 kg. Answer: 1500 kg.

4. The mass and momentum of a girl are 35 kg and 175 kg m s^-1 respectively. At what velocity is she travelling? Express your answer in km h^-1.

Step 1: v = p / m = 175 / 35 = 5 m s^-1. Step 2: convert: 5 x 3.6 = 18 km h^-1. Answers: 5 m s^-1, 18 km h^-1.

5. Calculate the momentum of a 60 kg player who is running with a velocity 7.5 m s^-1. If a 900 kg car moves with the same momentum as that of the player, then determine the velocity of car?

Step 1: player's momentum = 60 x 7.5 = 450 kg m s^-1. Step 2: car's velocity = p / m = 450 / 900 = 0.5 m s^-1. Answers: 450 kg m s^-1, 0.5 m s^-1.

6. A rifle has a mass of 5.0 kg. It fires a bullet of 8 g with a speed of 400 m s^-1. Calculate the recoil speed of the gun.

Step 1: bullet mass = 8 g = 0.008 kg; bullet momentum = 0.008 x 400 = 3.2 kg m s^-1. Step 2: by conservation of momentum, gun momentum = -3.2, so recoil speed = 3.2 / 5.0 = 0.64 m s^-1 in the opposite direction. Answer: -0.64 m s^-1 (0.64 m s^-1 backwards).

7. An object of mass 500 kg moves with a velocity 300 m s^-1. It splits into two parts. One of them has mass 100 kg and moves with a velocity 40 m s^-1 in the opposite direction while the second part of the object continues its motion in the same direction. Calculate the velocity of the second part of the object.

Step 1: total momentum before = 500 x 300 = 150,000 kg m s^-1. Step 2: first part's momentum = 100 x (-40) = -4,000 kg m s^-1. Step 3: second part's mass = 500 - 100 = 400 kg, so its velocity = (150,000 - (-4,000)) / 400 = 154,000 / 400 = 385 m s^-1. Answer: 385 m s^-1.

8. The mass of a cannon is 1000 kg. It fires a 20 kg cannon ball and recoils with a velocity of 0.6 m s^-1. What is the velocity of the cannon ball when it leaves the cannon?

Step 1: cannon's momentum = 1000 x 0.6 = 600 kg m s^-1 backwards. Step 2: ball's momentum must be 600 forwards, so ball velocity = 600 / 20 = 30 m s^-1. Answer: 30 m s^-1.

9. An object of mass 50 kg is moving with a velocity of 10 m s^-1. A constant force is applied on the object which increases its velocity to 20 m s^-1 in 5 seconds. Calculate the magnitude of applied force.

Step 1: change in momentum = 50 x (20 - 10) = 500 kg m s^-1. Step 2: F = change in momentum / time = 500 / 5 = 100 N. Answer: 100 N.

10. An object of mass 250 g is moving at a velocity of 64.8 km h^-1. What is initial momentum of the object? Calculate the value of impact time if a force of 9 kN is needed to stop the object.

Step 1: convert: mass = 0.25 kg, velocity = 64.8 / 3.6 = 18 m s^-1. Step 2: initial momentum = 0.25 x 18 = 4.5 kg m s^-1. Step 3: impulse to stop = 4.5 = F x (delta t), so (delta t) = 4.5 / 9000 = 5.0 x 10^-4 s. Answers: 4.5 kg m s^-1, 5.0 x 10^-4 s.

Unit 6: Turning Effect of Forces

Questions 6

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) Forces which are parallel to each other and acting in the same direction are called:

(a) like parallel force ✓

(b) unlike parallel forces

(c) perpendicular forces

(d) none of these

(ii) Forces which are parallel to each other and acting in the opposite directions are called:

(a) like parallel force

(b) unlike parallel forces ✓

(c) perpendicular forces

(d) none of these

(iii) The perpendicular distance between the axis of rotation of a body and the line of action of force applied on it is called:

(a) torque

(b) moment arm ✓

(c) tension

(d) momentum

(iv) Torque is defined as the product of:

(a) mass and acceleration

(b) mass and velocity

(c) force and displacement

(d) force and moment arm ✓

(v) The anticlockwise torque is taken as:

(a) positive ✓

(b) negative

(c) zero

(d) none of these

(vi) The clockwise torque is taken as:

(a) positive

(b) negative ✓

(c) zero

(d) none of these

(vii) The SI unit of moment of force or torque is

(a) N m ✓

(b) N m^-1

(c) N s

(d) N s^-1

(viii) An object having a lower centre of gravity is:

(a) less stable

(b) more stable ✓

(c) not stable

(d) none of these

(ix) An object having a higher centre of gravity is:

(a) less stable ✓

(b) more stable

(c) not stable

(d) none of these

(x) An object is more stable if it has:

(a) a lower centre of gravity

(b) a large base area

(c) both of these ✓

(d) none of these

(xi) The stability of an object can be improved by making:

(a) a lower centre of gravity

(b) a higher centre of gravity

(c) a large base area

(d) both (a) and (c) ✓

B. Give brief answers to the following questions.

(i) What is meant by parallel forces?

Parallel forces are forces whose lines of action are parallel to each other, whether they act in the same direction or in opposite directions. For example, the two forces of a couple acting on a steering wheel are parallel forces.

(ii) Differentiate between like and unlike parallel forces.

Like parallel forces are parallel forces acting in the same direction, so their resultant is the sum of the forces. Unlike parallel forces are parallel forces acting in opposite directions, so their resultant is the difference of the forces, acting in the direction of the larger force.

(iii) Define the term moment of force or torque.

The moment of force, or torque, is the turning effect of a force about a point or axis. It is defined as the product of the force and the perpendicular distance (moment arm) from the point of rotation to the line of action of the force: torque = F x d. Its SI unit is newton metre

(N m).

(iv) Give any two examples of moment of force from daily life.

Examples of moment of force: (1) opening a door by pushing its handle, far from the hinges; (2) loosening a nut with a spanner; (3) turning a steering wheel; (4) a see-saw balanced on its pivot.

(v) Explain why a long spanner is preferred to loosen or tighten a nut.

Torque = force x moment arm. With a long spanner the moment arm is larger, so the same force produces a larger torque, and a tight nut can be loosened with less effort. A short spanner would need a much larger force for the same torque.

(vi) Differentiate between positive and negative torques.

By convention, a torque that tends to turn a body in the anticlockwise direction is taken as positive, and a torque that tends to turn it in the clockwise direction is taken as negative. A body is in rotational equilibrium when the sum of positive torques equals the sum of negative torques.

(vii) State and explain the principle of moments.

The principle of moments states: a body will be at rest (in rotational equilibrium) if the sum of all clockwise torques about any point is equal to the sum of all anticlockwise torques about that point. For a balanced see-saw, for example, the clockwise moment on one side equals the anticlockwise moment on the other.

(viii) Define centre of gravity of an object. Give any two examples.

The centre of gravity of a body is the point at which the whole weight of the body appears to act vertically downwards towards the centre of the Earth. Examples: the centre of gravity of a uniform rod is at its midpoint; the centre of gravity of a uniform disc is at its centre; a racing car is designed with a low centre of gravity.

(ix) Describe briefly how position of centre of gravity affects the stability of an object.

The lower the centre of gravity of an object, the more stable the object: it is harder to topple, because tilting it raises its centre of gravity only slightly and the vertical line through it stays inside its base. A high centre of gravity makes an object less stable, since a small tilt moves that vertical line outside the base and the object topples over.

(x) Explain how the stability of an object can be improved.

The stability of an object can be improved in two ways: (1) by lowering its centre of gravity, for example by loading heavy luggage at the bottom of a bus; (2) by increasing its base area, for example by making a racing car wide and low. Both keep the vertical line through the centre of gravity inside the base for larger tilts.

(xi) What are applications of stability in real life? Give any two examples.

Applications of stability: (1) intercity buses carry luggage at the bottom, lowering the centre of gravity so the bus does not topple on turns; (2) racing cars are built low and wide so they

can take curves at high speed without toppling; (3) balancing toys are made so their centre of gravity stays below the pivot point, so they always return upright.

(xii) Describe briefly the motion of a rotating object.

A rotating object turns about a fixed point or axis, with all its particles moving in circles around that axis. In the absence of a net external torque it keeps rotating at the same rate (the rotational form of Newton's first law); a net torque in the direction of rotation speeds it up, and a net torque against it slows it down, until friction finally brings it to rest.

C. Numerical Problems (every step shown).

1. A force of 150 N is applied perpendicularly on a spanner at a distance of 20 cm from the nut. Calculate the torque produced by the force.

Step 1: convert distance: 20 cm = 0.20 m. Step 2: torque = F x d = 150 x 0.20 = 30 N m. Answer: 30 N m. (The book prints the unit as N s; the correct unit of torque is N m.)

2. A force of 50 N acts on an object. If the moment arm is 3 m, calculate the magnitude of torque.

Step 1: torque = F x d. Step 2: torque = 50 x 3 = 150 N m. Answer: 150 N m.

3. A door is 2 m wide and requires a torque of 80 N m to open it. Calculate the minimum value of force needed to open the door.

Step 1: F = torque / d. Step 2: F = 80 / 2 = 40 N. (Minimum force is obtained by pushing at the far edge, perpendicular to the door.) Answer: 40 N.

4. A force of 100 N is applied at the end of a spanner to open a nut. If the resulting torque is 50 N m, calculate the length of spanner.

Step 1: d = torque / F. Step 2: d = 50 / 100 = 0.5 m. Answer: 0.5 m.

5. A torque of 145 N m is needed to loosen a nut. If the length of spanner is 50 cm and the applied force at its edge is 300 N, calculate the angle between spanner and applied force.

Step 1: torque = F x d x sin(theta), with d = 50 cm = 0.5 m. Step 2: 145 = 300 x 0.5 x sin(theta) = 150 x sin(theta). Step 3: sin(theta) = 145 / 150 = 0.9667, so theta = 75 degrees. Answer: 75 degrees.

6. A uniform rod AB is balanced at point C. If a force of 20 N is applied at point A, then find the force 'F' to be applied to balance the rod. (AC = 10 m, CB = 5 m; both forces act downwards.)

Step 1: by the principle of moments about C, clockwise moment = anticlockwise moment. Step 2: 20 x 10 = F x 5. Step 3: F = 200 / 5 = 40 N. Answer: 40 N.

Unit 7: Centripetal Force

Questions 7

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) An object is moving in a circle. Which of the following quantities may remain constant during its motion?

(a) speed ✓

(b) velocity

(c) acceleration

(d) none of these

(ii) A force that keeps the body to move in the circle is called:

(a) centripetal force ✓

(b) centrifugal force

(c) gravitational force

(d) electrostatic force

(iii) The centripetal force is given by the formula:

(a) mv / r

(b) mv / r^2

(c) mv^2 / r ✓

(d) mv^2 / r

(iv) The centripetal force is directed:

(a) towards the centre of circle ✓

(b) away from the centre of circle

(c) tangent to the radius of circle

(d) none of these

(v) More centripetal force is needed if:

(a) speed of the body is decreased

(b) mass of the body is decreased

(c) mass of the body is increased ✓

(d) radius of the circle is increased

(vi) The magnitude of centripetal force will increase if we:

(a) increase the mass of the body

(b) increase the speed of the body

(c) decrease the radius of the circle

(d) all of these ✓

(vii) The magnitude of centripetal force is directly proportional to the:

(a) speed of the body

(b) square of speed of the body ✓

(c) radius of circle

(d) square of the radius of circle

(viii) The centrifugal force is directed:

(a) towards the centre of circle

(b) away from the centre of circle ✓

(c) tangent to the radius of circle

(d) none of these

(ix) The centripetal and centrifugal forces act:

(a) in the same direction

(b) in opposite directions ✓

(c) perpendicularly

(d) none of these

B. Give brief answers to the following questions.

(i) Explain briefly circular motion.

Circular motion is the motion of an object along a circular path. Its speed may stay constant, but its velocity keeps changing because its direction changes at every point, so it is accelerated motion even at constant speed.

(ii) What is meant by centripetal force?

Centripetal force is the force that keeps a body moving in a circle. It is always directed towards the centre of the circle, and without it the body would fly off along a tangent. The tension in a string, friction on a curved road, or gravity for a planet can all supply centripetal force.

(iii) What is the direction of centripetal force?

Centripetal force is always directed towards the centre of the circle, along the radius. It is perpendicular to the velocity of the body at every point, which is why it changes the direction of motion without changing the speed.

(iv) Write down the formula for centripetal force and explain the symbols used in the formula

Centripetal force Fc = m v^2 / r, where m is the mass of the body, v is its (constant) speed along the circle, and r is the radius of the circle. The corresponding centripetal acceleration is ac = v^2 / r, also towards the centre.

(v) Explain briefly centrifugal force.

Centrifugal force is a fictitious (apparent) force that appears to act on a body moving in a circle, equal in magnitude to the centripetal force but directed away from the centre, along the radius. It exists only as long as the body moves in the circle, and it is felt by the rotating body or the string, not as a real force on the body from outside.

(vi) What is the direction of the centrifugal force?

Centrifugal force is directed away from the centre of the circle, along the radius, opposite to the centripetal force. For example, passengers in a bus taking a turn are thrown outwards, away from the centre of the turn.

(vii) Give any two real life examples of centripetal force.

Real-life examples of centripetal force: (1) the tension in the string keeps a stone whirled in a circle moving in its path; (2) friction between tyres and road provides the centripetal force for a car taking a turn; (3) the Earth's gravity provides the centripetal force keeping the Moon in its orbit; (4) banking of roads helps provide the needed centripetal force safely.

C. Numerical Problems (every step shown).

1. A mass of 5 kg is kept moving in a circle of radius 2 m with a constant speed of 10 m s^-1. Find the centripetal force.

Step 1: Fc = m v^2 / r. Step 2: Fc = 5 x 10^2 / 2 = 5 x 100 / 2 = 250 N. Answer: 250 N.

2. Find the force required to keep a car of mass 2000 kg in an arc of radius 5 m when it is turning round a corner at 10 m s^-1.

Step 1: Fc = m v^2 / r. Step 2: Fc = 2000 x 10^2 / 5 = 2000 x 100 / 5 = 40,000 N = 4 x 10^4 N. Answer: 4 x 10^4 N.

3. The mass of the planet Venus is 4.87 x 10^24 kg and its average distance from the Sun is 1.08 x 10^11 m. If the planet Venus completes one revolution around the Sun in 225 days, calculate the centripetal force acting on the planet Venus.

Step 1: period T = 225 days = 225 x 24 x 3600 = 19,440,000 s = 1.944 x 10^7 s. Step 2: speed v = 2 pi r / T = 2 x 3.14 x 1.08 x 10^11 / 1.944 x 10^7 = 3.49 x 10^4 m s^-1. Step 3: Fc = m v^2 / r = 4.87 x 10^24 x (3.49 x 10^4)^2 / 1.08 x 10^11 = 5.5 x 10^22 N. Answer: about 5.5 x 10^22 N. (Note: the book's printed answer of 2.38 x 10^20 N is wrong by a factor of about 230; the value above follows from the book's own data.)

4. An aeroplane is moving in a circle of 25 km at constant height at speed 270 km h^-1. Calculate the centripetal acceleration acting on the aeroplane.

Step 1: convert: v = 270 / 3.6 = 75 m s^-1; r = 25 km = 25,000 m. Step 2: ac = v^2 / r = 75^2 / 25,000 = 5625 / 25,000 = 0.225 m s^-2. Answer: 0.225 m s^-2.

Unit 8: Work, Energy and Power

Questions 8

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) When the force and displacement are parallel to each other, then work is:

(a) minimum

(b) maximum ✓

(c) zero

(d) negative

(ii) The SI unit of work is:

(a) newton

(b) pascal

(c) watt

(d) joule ✓

(iii) The work done will be maximum if the angle between the force and the distance is:

(a) 0 degrees ✓

(b) 45 degrees

(c) 90 degrees

(d) 180 degrees

(iv) The work done will be zero if the angle between the force and the distance is:

(a) 0 degrees

(b) 45 degrees

(c) 90 degrees ✓

(d) 180 degrees

(v) Energy possessed by an object due to its motion is called its:

(a) gravitational potential energy

(b) elastic potential energy

(c) kinetic energy ✓

(d) chemical energy

(vi) The energy possessed by running water is called:

(a) gravitational potential energy

(b) elastic potential energy

(c) chemical energy

(d) kinetic energy ✓

(vii) If the velocity of a body is made twice, its kinetic energy will:

(a) remain the same

(b) become half

(c) become double

(d) become four times ✓

(viii) If the velocity of a body is made thrice, its kinetic energy will become:

(a) 3 times

(b) 6 times

(c) 9 times ✓

(d) 12 times

(ix) A car of mass 1000 kg is moving with a velocity of 30 m s^-1. Its kinetic energy will be:

(a) 3 x 10^5 J

(b) 6 x 10^5 J

(c) 9 x 10^5 J

(d) 4.5 x 10^5 J ✓

(x) The energy possessed by an object due to its position or configuration (shape or size) is called its:

(a) kinetic energy

(b) chemical energy

(c) heat energy

(d) potential energy ✓

(xi) When a ball is lifted from ground, then it will possess:

(a) kinetic energy

(b) potential energy ✓

(c) chemical energy

(d) electrical energy

(xii) Which types of energy is possessed by a compressed spring?

(a) Gravitational potential energy

(b) Elastic potential energy ✓

(c) Kinetic energy

(d) Wind energy

(xiii) One of the followings is a renewable source of energy?

(a) Coal

(b) Oil

(c) Gas

(d) Solar energy ✓

(xiv) One of the followings is a non-renewable source of energy?

(a) Solar energy

(b) Wind energy

(c) Coal ✓

(d) Geothermal energy

(xv) The rate of doing work with respect to time is called:

(a) energy

(b) power ✓

(c) momentum

(d) force

(xvi) The SI unit of power is:

(a) Joule

(b) Newton

(c) Coulomb

(d) Watt ✓

(xvii) The correct formula for power is:

(a) W t

(b) W t^2

(c) W / t ✓

(d) W / t^2

B. Give brief answers to the following questions.

(i) Define work. Give its SI unit.

Work is said to be done on an object when a force acts on it and produces a displacement in the direction of the applied force. Work = force x displacement in the direction of the force (W = Fd cos(theta)). Its SI unit is the joule (J): one joule is the work done when a force of one newton displaces an object through one metre in its own direction.

(ii) Define energy. Why do we need energy?

Energy is the capacity of a body to do work. We need energy for everything we do: moving, lifting, heating, lighting, running machines and vehicles, and carrying out every life process — without energy, no work and no activity is possible.

(iii) Differentiate between two types of mechanical energy.

The two types of mechanical energy are kinetic energy and potential energy. Kinetic energy is the energy a body possesses due to its motion (KE = 1/2 mv^2). Potential energy is the energy a body possesses due to its position or configuration, for example a ball held above the ground or a compressed spring.

(iv) Define kinetic energy and give its formula.

Kinetic energy is the energy possessed by an object due to its motion. Its formula is KE = (1/2) m v^2, where m is the mass and v is the velocity of the object. It is measured in joules and it is always positive.

(v) Can the value of kinetic energy be negative? Explain.

No, kinetic energy can never be negative. Since KE = (1/2) m v^2, mass is always positive

and v^2 is always positive (even if velocity is negative, its square is positive), so kinetic energy is always zero or positive; at rest it is zero.

(vi) Define potential energy.

Potential energy is the energy possessed by an object due to its position or configuration (shape or size). For example, water stored in a dam, a ball held above the ground, and a stretched or compressed spring all possess potential energy, which can be converted into kinetic energy.

(vii) What is meant by gravitational potential energy?

Gravitational potential energy is the energy possessed by an object due to its height above the surface of the Earth. It is given by PE = mgh, where m is the mass, g is the acceleration due to gravity and h is the height above the chosen reference level.

(viii) Differentiate between renewable and non-renewable sources of energy.

Renewable sources of energy can be used again and again or replenished in a short span of time, for example solar energy, wind energy, hydroelectric energy and geothermal energy. Non-renewable sources cannot be replenished in a short duration of time, for example coal, oil and natural gas (fossil fuels); once used up, they are gone for practical purposes.

(ix) Why fossil fuels are called non-renewable sources of energy?

Fossil fuels are called non-renewable because they were formed from dead plants and animals over millions of years under the Earth, and they are being used up far faster than nature can replace them. A source that cannot be replenished in a short duration of time is termed non-renewable.

(x) Name the major sources of energy.

The major sources of energy are: fossil fuels (coal, oil and natural gas), solar energy, wind energy, hydroelectric energy, geothermal energy, biomass and biogas, nuclear energy, and tidal and sea energy.

(xi) What is meant by hydroelectric resources?

Hydroelectric resources means the energy obtained from running or falling water. Water stored behind a dam is allowed to fall on turbines, and the kinetic energy of the moving water is converted into electrical energy by generators.

(xii) Give a few uses of solar panels.

Uses of solar panels: (1) producing electricity for homes, schools and street lights; (2) heating water in solar geysers; (3) running water pumps in fields; (4) charging batteries and running calculators and remote installations where grid electricity is not available.

(xiii) What are advantages and disadvantages of solar energy?

Advantages of solar energy: it is free, renewable, pollution-free and available almost everywhere. Disadvantages: it is available only in daytime and in fair weather, solar panels and storage batteries are costly, and a large area is needed for large-scale generation.

(xiv) State and explain wind energy.

Wind energy is the energy carried by blowing air. Moving air possesses kinetic energy, which windmills and wind turbines convert into mechanical energy and then into electrical energy. It is renewable and clean, but it depends on steady wind.

(xv) What is the main cause of tides in oceans?

The main cause of tides in oceans is the gravitational pull of the Moon on the Earth's oceans (with a smaller contribution from the Sun). The Moon's gravity pulls the ocean water towards it, creating a bulge (high tide), while inertia creates a second bulge on the opposite side.

(xvi) What is meant by the efficiency of a system?

The efficiency of a system is the ratio of the useful energy (or work) output to the total energy (or work) input, usually expressed as a percentage. It tells how much of the input energy is actually converted into useful form, the rest being lost, usually as heat.

(xvii) Give the formula for the efficiency of a system.

Efficiency = (useful output energy / total input energy) x 100%. Equivalently, efficiency = (useful work output / total work input) x 100%. Efficiency is always less than 100%, because some energy is always lost in every conversion.

(xviii) Is it possible to make a perpetual machine?

No, a perpetual machine — a machine that runs forever without any energy input and keeps giving out useful work — is not possible. In every energy conversion some energy is lost (usually as heat and friction), so the output is always less than the input, and the machine must eventually stop.

(xix) Explain the term power. Give its SI units.

Power is defined as the rate at which work is done or energy is consumed: power = work / time (P = W / t). Its SI unit is the watt (W); one horsepower (hp) = 746 W is another common unit.

(xx) Define one watt.

One watt is the power which does one joule of work in one second. So 1 W = 1 J s^-1: a machine rated at 1 watt converts or consumes one joule of energy every second.

C. Numerical Problems (every step shown).

1. A force of 150 N is applied on a box and the box covers a distance of 7.5 m in the direction of the force. Calculate the work done by the force.

Step 1: W = F x d (force and displacement in the same direction). Step 2: W = 150 x 7.5 = 1125 J. Answer: 1125 J.

2. A trolley of mass 5 kg is pulled by a force of 10 N. The trolley has travelled 4 m in the direction of the force. Find out (a) how much work has been done on it? (b) how much kinetic energy it has gained? (c) its velocity.

Step 1: work done W = F x d = 10 x 4 = 40 J. Step 2: by the work-energy principle, the kinetic energy gained equals the work done = 40 J. Step 3: KE = (1/2) m v^2, so 40 = 0.5 x

5 x v^2, giving v^2 = 16 and v = 4 m s^-1. Answers: (a) 40 J, (b) 40 J, (c) 4 m s^-1.

3. A body is pushed with a force of 500 N at an angle of 60 degrees with the horizontal direction. The body covers a distance of 10 m. How much work is done?

Step 1: W = F d cos(theta). Step 2: W = 500 x 10 x cos(60 degrees) = 5000 x 0.5 = 2500 J. Answer: 2500 J.

4. The mass of a girl is 45 kg. If she is running at 10 m s^-1 then calculate its kinetic energy.

Step 1: KE = (1/2) m v^2. Step 2: KE = 0.5 x 45 x 10^2 = 0.5 x 45 x 100 = 2250 J. Answer: 2250 J.

5. The mass of a car is 800 kg. Calculate the speed of car if its kinetic energy is 2.5 x 10^5 J.

Step 1: KE = (1/2) m v^2, so v^2 = 2 x KE / m. Step 2: v^2 = 2 x 2.5 x 10^5 / 800 = 500,000 / 800 = 625. Step 3: v = 25 m s^-1. Answer: 25 m s^-1.

6. An object has a mass of 25 kg. Calculate its potential energy. (a) 5 m above the ground (b) 10 m above the ground. Take the value of g as 9.8 m s^-2.

Step 1: PE = mgh. Step 2: (a) PE = 25 x 9.8 x 5 = 1225 J. Step 3: (b) PE = 25 x 9.8 x 10 = 2450 J. Answers: (a) 1225 J, (b) 2450 J.

7. A man lifts a parcel of mass 5 kg from the ground on to a shelf 2 m high. How much work he does on the parcel? What is the potential energy? Take the value of g as 9.8 m s^-2.

Step 1: work against gravity = mgh = 5 x 9.8 x 2 = 98 J, which the book rounds to 100 J. Step 2: the gain in potential energy equals the work done = 98 J (about 100 J). Answers: 100 J of work, 100 J of potential energy (98 J before rounding).

8. A man lifts a weight of 300 N through a vertical height of 2 m in 6 seconds. What power does he develop?

Step 1: work done W = F x h = 300 x 2 = 600 J. Step 2: power P = W / t = 600 / 6 = 100 W. Answer: 100 W.

9. A crane lifts a load weighing 3000 N through a height of 5 m in 10 seconds. What is the power of the crane in hp?

Step 1: work done W = 3000 x 5 = 15,000 J. Step 2: power P = 15,000 / 10 = 1500 W. Step 3: in hp, P = 1500 / 746 = 2.01, about 2 hp. Answer: 2 hp.

10. A quarter horse power motor is run for two hours. Determine the work done by the motor.

Step 1: power P = 0.25 hp = 0.25 x 746 = 186.5 W. Step 2: time t = 2 hours = 7200 s. Step 3: work W = P x t = 186.5 x 7200 = 1,342,800 J = 1.34 x 10^6 J. Answer: 1.34 x 10^6 J.

Unit 9: Pressure

Questions 9

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) Pressure is defined as:

(a) force x area

(b) force / area ✓

(c) area / force

(d) 1 / (force x area)

(ii) The pressure exerted by an object can be increased by:

(a) increasing applied force

(b) reducing the contact area

(c) both of these ✓

(d) none of these

(iii) The pressure increases with:

(a) decrease in applied force

(b) decrease in area ✓

(c) increase in area

(d) none of these

(iv) The SI unit of pressure is

(a) J

(b) N

(c) N m^-1

(d) N m^-2 ✓

(v) One atmospheric pressure is equal to pressure exerted by:

(a) 76 cm tall column of alcohol

(b) 76 cm tall column of water

(c) 76 cm tall column of glycerine

(d) 76 cm tall column of mercury ✓

(vi) The value of the atmospheric pressure at sea level is equal to:

(a) 0.76 mm of mercury column

(b) 7.6 mm of mercury column

(c) 76 mm of mercury column

(d) 760 mm of mercury column ✓

(vii) One atmospheric pressure is equal to:

(a) 1.013 x 10^3 N m^-2

(b) 1.013 x 10^4 N m^-2

(c) 1.013 x 10^5 N m^-2 ✓

(d) 1.013 x 10^6 N m^-2

(viii) Bar is a unit of:

(a) force

(b) area

(c) volume

(d) pressure ✓

(ix) As we move away from the surface of the Earth, the atmospheric pressure:

(a) remains constant

(b) increases

(c) decreases ✓

(d) none of these

(x) At an altitude of 30 km, the atmospheric pressure drops to:

(a) 7 mm of mercury

(b) 10 mm of mercury ✓

(c) 20 mm of mercury

(d) 25 mm of mercury

(xi) A rapid increase in atmospheric pressure indicates that there will be:

(a) rain

(b) storm

(c) poor weather

(d) pleasant weather ✓

(xii) Which of the following instrument is used to measure the atmospheric pressure?

(a) Vernier callipers

(b) Electronic balance

(c) Altimeter

(d) Barometer ✓

(xiii) The pressure at a point in a liquid is given by the relation:

(a) P = rho g / h

(b) P = rho h / g

(c) P = rho g h ✓

(d) P = 1 / (rho g h)

(xiv) The pressure exerted by a liquid depends on:

(a) depth of the liquid

(b) density of the liquid

(c) both of these ✓

(d) none of these

(xv) The pressure exerted by a liquid can be increased by:

(a) keeping constant depth

(b) decreasing the depth

(c) increasing the depth ✓

(d) none of these

(xvi) The use of Pascal's law is made in:

(a) hydraulic press

(b) hydraulic brake

(c) hydraulic jack

(d) all of these ✓

(xvii) Hydraulic press works on:

(a) Newton's first law of motion

(b) Hook's law

(c) Pascal's law ✓

(d) Archimedes principle

B. Give brief answers to the following questions.

(i) Define the term pressure. Give its SI unit.

Pressure is defined as the force per unit area applied perpendicularly on the surface of an object: P = F / A. Its SI unit is the newton per square metre (N m^-2), also called the pascal (Pa): 1 Pa = 1 N m^-2.

(ii) State the factors that affect the pressure.

Pressure depends on two factors: (1) the applied force — a larger force produces a larger pressure on the same area; (2) the area over which the force acts — a smaller area produces a larger pressure for the same force. For a liquid, pressure also depends on the depth and the density of the liquid.

(iii) Explain briefly the working of a rubber sucker.

A rubber sucker is pressed against a smooth surface so that the air between the sucker and the surface is pushed out, leaving almost no air inside. The atmospheric pressure acting on the outside of the sucker is then much greater than the pressure inside, and it holds the sucker firmly against the surface.

(iv) Why do we use a sharp knife to cut fruit and vegetables?

A sharp knife has a very small edge area. For the same applied force, pressure = force / area is much larger on the small edge, so the knife cuts easily. A blunt knife spreads the same force over a larger area and produces too little pressure to cut.

(v) Explain why broad tyres are used in a tractor.

Broad tyres spread the weight of the tractor over a larger contact area, so the pressure on the ground (force / area) is smaller. The tractor therefore does not sink into soft fields, and it grips better without damaging the soil.

(vi) When you walk on sand then your foot prints are left on sand. Explain.

When you walk on sand, your weight acts on the small area of your foot, producing a large pressure on the sand grains. The sand cannot support that pressure, so it is displaced and compressed, leaving a footprint shaped like your foot.

(vii) Define the term atmospheric pressure.

Atmospheric pressure is the force per unit area exerted on a surface by the weight of the air column above that surface in the atmosphere of the Earth. At sea level it is about 1.013 x 10^5 N m^-2, equal to the pressure of a 76 cm column of mercury.

(viii) Give at least two applications of the atmospheric pressure.

Applications of atmospheric pressure: (1) a rubber sucker sticks to a wall because outside air pressure holds it; (2) drinking through a straw works because sucking lowers the pressure inside the straw and atmospheric pressure pushes the liquid up; (3) a barometer measures atmospheric pressure to forecast weather; (4) a syringe fills when its piston is pulled back.

(ix) Why does the atmospheric pressure vary with altitude?

Atmospheric pressure is caused by the weight of the air above. As altitude increases, the column of air above becomes shorter and thinner, so its weight decreases and the pressure falls. That is why pressure at 30 km altitude is only about 10 mm of mercury, against 760 mm at sea level.

(x) How will you correlate the sudden increase in atmospheric pressure to weather?

A sudden (rapid) increase in atmospheric pressure indicates that dry, heavy air is settling, which generally brings clear, pleasant weather. A sudden fall in pressure indicates moist, rising air and usually brings rain, storm or poor weather.

(xi) What is a barometer?

A barometer is the device used to measure atmospheric pressure. The common mercury barometer is a long glass tube filled with mercury and inverted in a mercury reservoir: the height of the mercury column (about 76 cm at sea level) measures the atmospheric pressure.

(xii) Explain why water cannot be used as barometric liquid.

Water cannot be used as a barometric liquid because its density is about 13.6 times smaller than mercury, so a water barometer would need a column about 10.3 m tall to balance atmospheric pressure — far too tall for a practical instrument. Water also evaporates and sticks to glass, making readings unreliable.

(xiii) State the Pascal's law.

Pascal's law states: the pressure applied to a completely enclosed liquid is transmitted without any loss to all other parts of the liquid and to the enclosing walls. In other words, pressure acts equally in all directions in an enclosed liquid, which is the principle of

hydraulic machines.

(xiv) Describe briefly hydraulic jack.

A hydraulic jack works on Pascal's law. It has a small piston (pump) and a large piston connected by a liquid. A small force on the small piston creates a pressure that is transmitted equally to the large piston; because the large piston has a much larger area, it lifts a much larger load. Pumping the handle repeatedly raises heavy vehicles with little effort.

C. Numerical Problems (every step shown).

1. The length of a side of an iron cube is 20 cm. If the density of iron is 7900 kg m^-3, calculate; (a) volume of the cube (b) mass of the cube. (c) weight of the cube. (d) The cube is placed on the top surface of a table. What pressure will be exerted on table by the base of cube? Take the value of g as 9.8 m s^-2.

Step 1: side = 20 cm = 0.20 m. (a) Volume = 0.20^3 = 0.008 m^3 = 8 x 10^-3 m^3. Step 2: (b) mass = density x volume = 7900 x 0.008 = 63.2 kg. Step 3: (c) weight = mg = 63.2 x 9.8 = 619.36 N. Step 4: (d) base area = 0.20 x 0.20 = 0.04 m^2, so pressure = 619.36 / 0.04 = 15,484 Pa. Answers: (a) 8 x 10^-3 m^3, (b) 63.2 kg, (c) 619.36 N, (d) 15,484 Pa. (Note: the book prints (d) as 15484 N; the value is right but the unit must be pascal, N m^-2, not N.)

2. A hammer exerts a force of 2.5 N on a nail of tip area 1 mm^2. Calculate the pressure on the nail.

Step 1: convert area: 1 mm^2 = 1 x 10^-6 m^2. Step 2: pressure = F / A = 2.5 / (1 x 10^-6) = 2.5 x 10^6 N m^-2. Answer: 2.5 x 10^6 N m^-2.

3. The length, breadth and height of a rectangular wooden block are 60 cm, 30 cm and 15 cm respectively. The mass of this block is 13.5 kg and it is placed on the top of a table. Calculate the maximum and minimum values of pressure exerted by block on the top surface of the table. Take the value of g as 9.8 m s^-2.

Step 1: weight = 13.5 x 9.8 = 132.3 N. Step 2: the three face areas are 0.60 x 0.30 = 0.18 m^2, 0.60 x 0.15 = 0.09 m^2 and 0.30 x 0.15 = 0.045 m^2. Step 3: maximum pressure is on the smallest face: 132.3 / 0.045 = 2940 Pa. Step 4: minimum pressure is on the largest face: 132.3 / 0.18 = 735 Pa. Answers: maximum 2940 Pa, minimum 735 Pa. (Note: the book's printed maximum of 4940 N is wrong in both value and unit; the minimum 735 Pa is correct.)

4. The depth of a domestic underground water reservoir is 2.5 m. Calculate the pressure exerted by water on bottom of the reservoir if the density of water is 1 x 10^3 kg m^-3. What pressure will be exerted by water on the walls of the reservoir? The density of water is 1 x 10^3 kg m^-3.

Step 1: pressure at the bottom P = rho g h = 1000 x 9.8 x 2.5 = 24,500 Pa = 2.45 x 10^4

Pa. Step 2: pressure in a liquid acts equally in all directions, so at the same depth (at the bottom) the pressure on the walls is the same 2.45 x 10^4 Pa; higher up the wall the pressure is less, because it increases with depth. Answer: 2.45 x 10^4 Pa at the bottom, the same on the walls at that depth.

5. The length, width and height of a water tank are 8 m, 5 m and 1.5 m respectively. It is filled with water. (a) Calculate the pressure of water at the bottom of the tank. (b) What is force acting at the bottom of the tank? The density of water is 1 x 10^3 kg m^-3.

Step 1: (a) pressure at the bottom P = rho g h = 1000 x 9.8 x 1.5 = 14,700 Pa = 1.47 x 10^4 Pa. Step 2: (b) bottom area = 8 x 5 = 40 m^2, so force = P x A = 14,700 x 40 = 588,000 N = 5.88 x 10^5 N. Answers: (a) 1.47 x 10^4 N m^-2, (b) 5.88 x 10^5 N.

6. What will be the pressure exerted by 75 cm vertical column of alcohol if the density of alcohol is 0.8 g cm^-3?

Step 1: convert: density = 0.8 g cm^-3 = 800 kg m^-3; height = 75 cm = 0.75 m. Step 2: P = rho g h = 800 x 9.8 x 0.75 = 5880 Pa = 5.88 x 10^3 Pa. Answer: 5.88 x 10^3 Pa.

7. What height of a vertical column of water will exert a pressure of 4.9 x 10^5 N m^-2? The density of water is 1 x 10^3 kg m^-3.

Step 1: h = P / (rho g). Step 2: h = 4.9 x 10^5 / (1000 x 9.8) = 490,000 / 9800 = 50 m. Answer: 50 m.

8. The average weight of an elephant is 4000 N. The surface area of the sole of its four feet is 0.1 m^2. Calculate the pressure exerted by one foot of an elephant.

Step 1: the load carried by one foot is 4000 / 4 = 1000 N. Step 2: taking the book's reading that each foot's sole area is 0.1 m^2, pressure by one foot = 1000 / 0.1 = 10,000 Pa. Answer: 10,000 Pa. (Note: the question's wording is ambiguous — if 0.1 m^2 is the total area of all four feet, each foot has 0.025 m^2 and the pressure per foot is 40,000 Pa; the book's answer follows the per-foot reading.)

Unit 10: Deformation of Solids

Questions 10

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) Forces can produce a change in _____ of an object.

(a) length

(b) area

(c) volume

(d) all of these ✓

(ii) Which of the following expressions represent the Hooke's law?

(a) F = kx ✓

(b) F = k / x

(c) F = x / k

(d) F = 1 / (kx)

(iii) A stiffer spring has a _____ value of its spring constant.

(a) zero

(b) small

(c) medium

(d) large ✓

(iv) Hooke's law is applicable to:

(a) solids ✓

(b) liquids

(c) gases

(d) all of these

(v) If the applied force on a spring is within elastic limit, then load-extension graph obeys:

(a) Newton's first law of motion

(b) Newton's second law of motion

(c) Newton's third law of motion

(d) Hooke's law ✓

(vi) The shape of load-extension graph is:

(a) a straight line parallel to x-axis

(b) a straight line parallel to y-axis

(c) an inclined straight line ✓

(d) none of these

(vii) Which of the following devices work on Hooke's law?

(a) Spring mattress

(b) Suspension system of a car

(c) a spring balance or scale

(d) All of these ✓

B. Give brief answers to the following questions.

(i) Describe briefly the changes produced in an object by the applied forces.

A force applied to a solid object can produce a change in its size — its length, area or volume — and in its shape. This change in size and shape is called deformation of the solid. If the force is within the elastic limit, the object returns to its original size and shape when the force is removed; beyond it, the deformation becomes permanent.

(ii) State and briefly explain Hooke's law.

Hooke's law states: the force needed to stretch or compress a spring by some distance is proportional to that distance, provided the applied force does not exceed the elastic limit of the spring. In symbols, F = kx, where F is the deforming force, x is the extension (or compression) and k is the spring constant.

(iii) What is meant by the term spring constant?

The spring constant (k) is the ratio of the deforming force to the resulting extension in the spring: k = F / x. It gives a direct measure of the stiffness of the spring — a stiffer spring has a larger spring constant — and its unit is newton per metre (N m^-1).

(iv) What are the limitations of Hooke's law?

Hooke's law holds only within the elastic limit (limit of proportionality) of the material. Beyond that limit the extension is no longer proportional to the force, the material does not return fully to its original shape, and eventually it may break. It also applies to solids, not to liquids or gases.

(v) Define the term 'limit of proportionality' or elastic limit for a load-extension graph.

The limit of proportionality for a load-extension graph is the limit up to which Hooke's law is valid — the point up to which extension is directly proportional to load and the graph is a straight line through the origin. Beyond the elastic limit the spring does not fully recover its original length when the load is removed.

(vi) Describe briefly load-extension graph for a spring.

For a spring within its elastic limit, the load-extension graph is an inclined straight line through the origin: extension increases in direct proportion to load, verifying Hooke's law. The slope of the line (load / extension) gives the spring constant k. Beyond the limit of proportionality the graph curves away from the straight line.

(vii) Give any two applications of Hooke's law.

Applications of Hooke's law: (1) a spring balance or scale measures weight or mass, because the extension of its spring is proportional to the load; (2) vehicle suspension systems and shock absorbers use springs chosen by their spring constant; (3) spring

mattresses, the balance wheel of a mechanical watch, and moving-coil instruments such as galvanometers, ammeters and voltmeters all work on Hooke's law.

C. Numerical Problems (every step shown).

1. A spring hangs vertically with its upper end fixed to a clamp of an iron stand. A weight of 10 N is attached to its lower end and an extension of 5 mm in the spring is observed. Calculate the value of spring constant.

Step 1: convert extension: 5 mm = 0.005 m. Step 2: k = F / x = 10 / 0.005 = 2000 N m^-1. Answer: 2000 N m^-1.

2. A spring has a spring constant 200 N m^-1. How much force will be needed to produce 25 cm extension in the given spring?

Step 1: convert extension: 25 cm = 0.25 m. Step 2: F = kx = 200 x 0.25 = 50 N. Answer: 50 N.

3. A spring having spring constant 300 N m^-1 hangs vertically from a rigid support. What will be the extension produced in the spring if a 6 N weight is attached at its lower end?

Step 1: x = F / k. Step 2: x = 6 / 300 = 0.02 m = 20 mm. Answer: 20 mm.

4. In an experiment with a spring, the following data was obtained. Load in newtons: 0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0. Length of spring in mm: 200, 225, 250, 275, 300, 325, 350. (a) Complete third row of the above table (extension). (b) Plot a load-extension graph by taking load along the x-axis and extension along the y-axis. (c) Comment on the shape of the graph. (d) Calculate the spring constant.

(a) Extension = length - original length (200 mm): 0, 25, 50, 75, 100, 125, 150 mm. (b) Plot load on the x-axis and extension on the y-axis: the points (0, 0), (0.5, 25), (1.0, 50), (1.5, 75), (2.0, 100), (2.5, 125), (3.0, 150) lie on one line. (c) The graph is a straight line through the origin, showing extension is directly proportional to load — Hooke's law is obeyed within this range. (d) Spring constant k = load / extension = 0.5 N / 25 mm = 0.5 / 0.025 = 20 N m^-1 (the same from any pair, e.g. 3.0 / 0.150 = 20 N m^-1). Answer: 20 N m^-1.

Unit 11: Density

Questions 11

A. Multiple choice questions. (Correct option marked with a green tick.)

(i) The density of a substance is given by:

(a) mass x volume

(b) mass / volume ✓

(c) volume / mass

(d) 1 / (mass x volume)

(ii) The SI unit of density is equal to:

(a) kg m^-1

(b) kg m^-2

(c) kg m^-3 ✓

(d) none of these

(iii) Density is also expressed as:

(a) g cm^-1

(b) g^-1 cm

(c) g cm^-2

(d) g cm^-3 ✓

(iv) The SI unit of volume is:

(a) (1 / m^3)

(b) mm^3

(c) cm^3

(d) m^3 ✓

(v) Mass of an object is given by:

(a) density / volume

(b) volume / density

(c) density x volume ✓

(d) 1 / (density x volume)

(vi) If 'a' represents the side of a cube then its volume is given by:

(a) a

(b) a^2

(c) a^3 ✓

(d) (1 / a^3)

(vii) If 'r' represents the radius of a solid sphere the its volume is given by:

(a) 2 pi r

(b) pi r^2

(c) (4/3) pi r^3 ✓

(d) pi r^2

(viii) If 'r' and 'h' represent the radius and height of a solid cylinder respectively then its volume is given by:

(a) 2 pi r^2 h

(b) 2 pi r h

(c) pi r h^2

(d) pi r^2 h ✓

B. Give brief answers to the following questions.

(i) Define the term density.

Density of a substance is defined as its mass per unit volume: density = mass / volume (rho = m / V). It tells how much mass is packed into a given volume of the substance.

(ii) What are SI units of density?

The SI unit of density is kilogram per cubic metre (kg m^-3). Density is also commonly expressed in gram per cubic centimetre (g cm^-3), and 1 g cm^-3 = 1000 kg m^-3.

(iii) Under what condition an object will float on the surface of the liquid?

An object will float on the surface of a liquid if its density is less than the density of the liquid. If its density is greater than the liquid's density, it will sink; if the densities are equal, it stays suspended inside the liquid.

(iv) Explain why ice floats on the surface of water.

Ice floats on water because the density of ice (about 0.92 g cm^-3) is less than the density of water (1.0 g cm^-3). When water freezes, it expands, so the same mass occupies a larger volume and its density falls below that of liquid water.

(v) A piece of iron floats on the surface of mercury but sinks in water. Explain.

Iron has a density of about 7.8 g cm^-3. Mercury has a much larger density (about 13.6 g cm^-3), so iron is less dense than mercury and floats on it. Water has a density of only 1.0 g cm^-3, so iron is denser than water and sinks in it.

(vi) Give the formula for volume of a solid sphere.

The volume V of a solid sphere of radius r is V = (4/3) pi r^3, where pi is about 3.14. For example, a sphere of radius 2 m has volume (4/3) x 3.14 x 8 = 33.5 m^3.

(vii) Give the formula for volume of a cylinder.

The volume V of a solid cylinder of radius r and height h is V = pi r^2 h, where pi is about 3.14. It is the area of the circular base (pi r^2) multiplied by the height.

(viii) Describe briefly how the density of an irregularly shaped solid can be measured.

Weigh the solid on a balance to find its mass m. Then find its volume by the displacement method: pour water into a measuring cylinder, note the level, lower the solid fully into the water, and note the new level — the rise in level equals the volume V of the solid. Density is

then rho = m / V.

C. Numerical Problems (every step shown).

1. A metallic object of mass 81 g has a volume of 30 cm^3. Calculate the density of the object in SI units. Can you identify the metal?

Step 1: density = mass / volume = 81 / 30 = 2.7 g cm^-3. Step 2: convert: 2.7 g cm^-3 = 2.7 x 1000 = 2700 kg m^-3. Step 3: this is the density of aluminium. Answers: 2700 kg m^-3, Aluminium.

2. The mass and radius of the Earth are 6 x 10^24 kg and 6.27 x 10^6 m respectively. Calculate the average density of the Earth.

Step 1: volume of Earth (as a sphere) V = (4/3) pi R^3 = (4/3) x 3.14 x (6.27 x 10^6)^3 = 4.19 x 2.46 x 10^20 = 1.03 x 10^21 m^3. Step 2: average density = mass / volume = 6 x 10^24 / 1.03 x 10^21 = 5.81 x 10^3 kg m^-3. Answer: about 5.81 x 10^3 kg m^-3. (Note: the book prints 5.56 x 10^3 kg m^-3, which corresponds to a radius of about 6.37 x 10^6 m, not the printed 6.27 x 10^6 m; the value above uses the book's printed data.)

3. A cube is made from expanded polystyrene which has a density 16 kg m^-3. If the mass of this cube is 128 g then calculate the side of cube.

Step 1: convert mass: 128 g = 0.128 kg. Step 2: volume = mass / density = 0.128 / 16 = 0.008 m^3. Step 3: side = cube root of volume = cube root of 0.008 = 0.2 m. Answer: 0.2 m, or 20 cm.

4. The length, width and height of a rectangular block of ice are 150 cm, 50 cm and 25 cm respectively. If the density of ice is 0.92 g cm^-3, calculate the mass of block of ice in kg.

Step 1: volume = 150 x 50 x 25 = 187,500 cm^3. Step 2: mass = density x volume = 0.92 x 187,500 = 172,500 g. Step 3: convert: 172,500 g = 172.5 kg. Answer: 172.5 kg.

5. The diameter of a stainless steel ball bearing is measured as 4.344 mm with a digital screw gauge. (a) Calculate the volume of steel ball bearing with reasonable number of significant numbers. (b) Convert the volume from mm^3 into m^3. (c) If the mass of given ball bearing measured by electronic balance is 334.62 g then calculate the density of stainless steel.

Step 1: radius = diameter / 2 = 4.344 / 2 = 2.172 mm. Step 2: (a) volume = (4/3) pi r^3 = 4.19 x (2.172)^3 = 4.19 x 10.25 = 42.9 mm^3 (three significant figures, matching the measurement). Step 3: (b) 1 mm^3 = 10^-9 m^3, so volume = 42.9 x 10^-9 = 4.29 x 10^-8 m^3. Step 4: (c) density = 334.62 g / 42.9 mm^3 = 7.80 g cm^-3 = 7800 kg m^-3. Answers: (a) 42.9 mm^3, (b) 4.29 x 10^-8 m^3, (c) 7800 kg m^-3. (Note: the book prints (b) as 4.29 x 10^-5 m^3, which is wrong by a factor of 1000 — 1 mm^3 = 10^-9 m^3, not 10^-6 m^3.)

6. A small cylinder is made from pinewood which has a density 500 kg m^-3. A Vernier callipers was used to take the following readings. Length of cylinder =

25.6 mm. Diameter of cylinder = 3.2 mm. (a) Calculate the volume of cylinder in mm^3 with reasonable number of significant numbers. (b) Convert the volume from mm^3 into m^3. (c) Calculate the mass of this wooden cylinder.

Step 1: radius = 3.2 / 2 = 1.6 mm. Step 2: (a) volume = pi r^2 h = 3.14 x (1.6)^2 x 25.6 = 3.14 x 2.56 x 25.6 = 206 mm^3. Step 3: (b) volume = 206 x 10^-9 = 2.06 x 10^-7 m^3. Step 4: (c) mass = density x volume = 500 x 2.06 x 10^-7 = 1.03 x 10^-4 kg = 0.103 g. Answers: (a) 206 mm^3, (b) 2.06 x 10^-7 m^3, (c) 1.03 x 10^-4 kg (0.103 g). (Note: the book's printed (b) has garbled units and a wrong exponent, and its (c) of 0.103 kg is 1000 times too large — it follows only if the volume were 206 cm^3 instead of 206 mm^3.)

7. A hollow metallic cylinder is made from iron and is shown in the figure. The following measurements were made using a Vernier callipers. Internal diameter of hollow cylinder = d = 4.12 cm. External diameter of hollow cylinder = D = 8.68 cm. Height of hollow cylinder = h = 12.75 cm. (a) Calculate the volume of hollow cylinder in cm^3 with reasonable number of significant numbers. (b) Express this value of volume in m^3. (c) Calculate the mass of hollow cylinder. Given that density of iron is 11600 kg m^-3.

Step 1: outer radius R = 8.68 / 2 = 4.34 cm; inner radius r = 4.12 / 2 = 2.06 cm. Step 2: (a) volume = pi h (R^2 - r^2) = 3.14 x 12.75 x (4.34^2 - 2.06^2) = 3.14 x 12.75 x (18.84 - 4.24) = 3.14 x 12.75 x 14.59 = 584 cm^3. Step 3: (b) volume = 584 x 10^-6 = 5.84 x 10^-4 m^3. Step 4: (c) mass = 11600 x 5.84 x 10^-4 = 6.78 kg. Answers: (a) 584 cm^3, (b) 5.84 x 10^-4 m^3, (c) 6.78 kg. (Note: the book prints (a) 591 cm^3 and (c) 6.86 kg, which follow from each other but not from the printed diameters; the book's iron density of 11600 kg m^-3 is also much larger than the usual value of about 7900 kg m^-3, but it is used as printed for (c).)

8. The mass of an empty beaker is measured on an electronic balance as 150.7 g. When 100 ml of cooking oil is poured into the beaker then reading on display of electronic balance becomes 222.5 g. Calculate the density of cooking oil.

Step 1: mass of oil = 222.5 - 150.7 = 71.8 g. Step 2: volume of oil = 100 ml = 100 cm^3. Step 3: density = 71.8 / 100 = 0.718 g cm^-3 = 718 kg m^-3. Answer: 718 kg m^-3.

Units 12-14: FLAGGED UNOBTAINABLE

Unit 12: Particle Theory of Matter (contents book p.232). The source file contains only the opening lesson pages (book pp.232-234) and then ends (p.235 is blank). Its end-of-unit Questions section is not in the file, so no Unit 12 questions are solved here — nothing has been invented for this unit.

Unit 13: Magnetism (contents book p.250). Not in the source file at all. No questions are solved here.

Unit 14: Space Physics (contents book p.287), Bibliography (p.306) and Glossary (p.307). Not in the source file at all. No questions are solved here.

This flagged-partial state is by plan (recovery of a complete source was closed as NOT FOUND on 2026-10-26). If a complete BTBB Grade 9 Physics file becomes available, Units 12-14 must be added to this PDF before it is described as complete.